

Natural products against cisplatin-induced male reproductive toxicity: A comprehensive review

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Abstract

Cisplatin is widely used as one of the most effective anticancer agents in the treatment of some neoplasms. Reproductive toxicity is the most common outcome associated with cisplatin testicular damage. Alternative natural medicines for treating male testicular disorders and infertility have received extensive attention in research. Natural products, medicinal herbs, and their secondary metabolites have been shown as promising agents in the management of testicular damage induced by chemotherapy drugs. This study aimed to review the research related to natural substances that are promising in mitigation of the cisplatin-induced toxicity in the reproductive system. PubMed and Scopus were searched for studies on various natural products for their potential protective property against reproductive toxicity induced by cisplatin from 2000 to 2020. Eligibility was checked based on selection criteria. Fifty-nine articles were included in this review. Mainly in animal studies, several natural agents have positively affected cisplatin-reproductive-toxicity factors, including reactive oxygen species, inflammatory mediators, DNA damage, and activation of the mitochondrial apoptotic pathway. Most of the natural agents were investigated in short-term duration and high doses of cisplatin exposure, considering their antioxidant activity against oxidative stress. Considering antioxidant properties, various natural products might be effective for the management of cisplatin reproductive toxicity. However, long-term recovery of spermatogenesis and management of low-dose-cisplatin toxicity should be considered as well as the bioavailability of these agents before and after treatment with cisplatin without affecting its anticancer activity.

KEYWORDS

chemotherapy, herbal medicine, phytochemicals, sperm, testicular damage

1 | INTRODUCTION

Generally, infertility and reproductive problems are among the complications occurring after chemotherapy and radiotherapy in cancer patients.^[1] Cisplatin (CP), one of the most effective anticancer agents, is widely used in the treatment of solid neoplasms such as testicular cancer.^[2] Infertility is the most common

outcome associated with CP-induced testicular damage.^[3] It has been reported that the adverse effects of CP on the testes are characterized by various anomalies such as atrophic seminiferous tubules, loss of connections between the spermatogonium and basal lamina, apoptosis, and necrosis in germinal epithelium, inflammation, and damage in the peritubular area which all lead to infertility.^[4]

CP disrupts nucleic acid synthesis in germ cells, which consequently disturbs the entire process of spermatogenesis. Also, testosterone production is impaired due to the fact that Leydig cells are damaged by CP.^[2] It creates DNA adducts and disrupts vital processes such as replication and transcription, and ultimately leads to apoptosis.^[2] Reproductive toxicity of CP seems to be due to the production of reactive oxygen species (ROS), which damage the testicular cells especially the spermatozoa. The toxicity is manifested through concomitant depletion of testicular antioxidant capacity and increase in lipid peroxidation.^[5] Need to mention that ROS generation depends on the dose of CP and the duration of treatment.^[6] The primary mechanism of killing proliferating cells by CP is forming DNA adducts that can cross-link the DNA. Proliferating spermatogonia can be killed at low doses of CP, likely by the cross-links.^[7]

Assisted reproduction and fertility preservation techniques that are used for cancer patients have their own shortcomings such as the decline in semen quality in men^[8] and relatively low success rate of the procedures.^[9] Therefore, alternative methods of treating male infertility have received wide attention in research areas these days. Natural products, medicinal herbs, and their secondary metabolites have been shown to have potential effectiveness in the management of infertility. The biological and pharmacological effects of these agents have been frequently examined in numerous research studies especially in the field of infertility and reproductive system disorders.

Considering that CP toxicity might be ascribed to oxidative stress, natural antioxidants may have the potentials to prevent toxicity of CP in organs especially the male reproductive system.^[10] It is shown that well-known antioxidants such as vitamin C are capable of

reducing many CP-associated toxicities. Other natural products have also been demonstrated to have protective activity against CP-induced toxicity for instance curcumin,^[11] resveratrol,^[12] melatonin,^[13] as well as several herbs such as *Zingiber officinale* Roscoe,^[14] *G. biloba* L.^[15] or organic agents like amifostine.^[16] However, these results have not been proven by clinical studies which necessitate further investigations to develop testicular protective agents against CP toxicity without affecting its anticancer activity.

The purpose of this study is to review available studies on the natural products' effects on CP-induced reproductive toxicity. In this article, we have reviewed the research related to natural products that are shown to be promising in mitigation of the CP-induced toxicity in the reproductive system. Electronic databases, including PubMed and Scopus, were searched for articles on various natural products for their potential protective property against reproductive toxicity induced by CP from 2000 to the end of 2020. All the authors performed the literature review and relevant data collection. As inclusion criteria, the data were collected from only original research articles published in the English language, with available full text. Studies irrelevant to reproductive toxicity and studies on the synthetic derivatives of phytochemicals were excluded. The search syntax included cisplatin AND (natural OR plant OR extract OR herb OR phytochemical) AND (testicular OR sperm OR testis OR epididymis OR reproductive OR urogenital) AND (toxicity OR damage OR injury). Every researcher independently conducted a literature evaluation as well as data extraction of each study that met the inclusion criteria. Of the initial 228 papers, 59 papers finally were included in this review after excluding duplicates, paper type, and irrelevant studies as shown in a flowchart of the selection process (Figure 1).

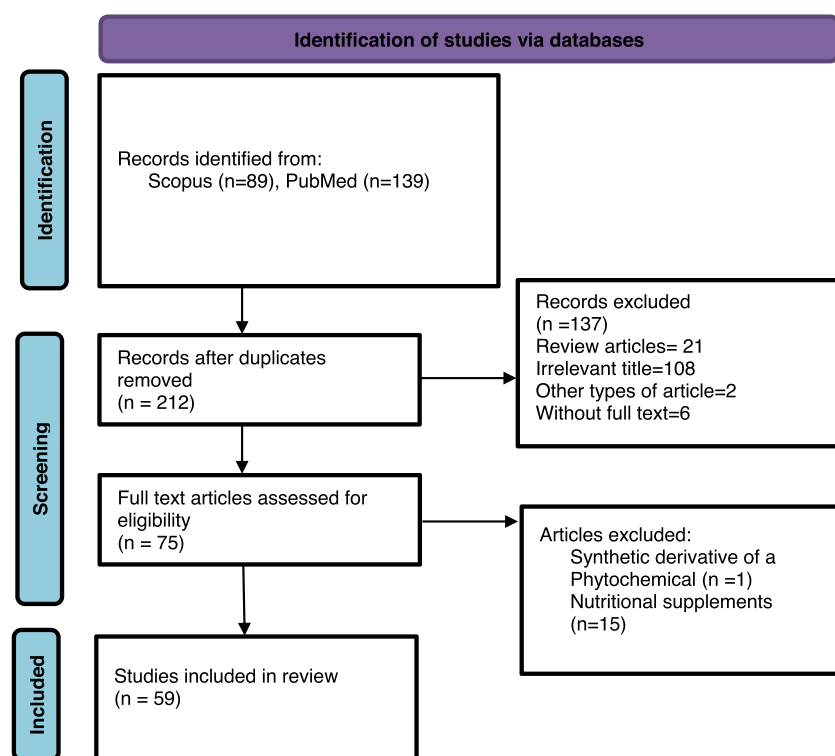


FIGURE 1 Flowchart of the selection process

The obtained data were classified based on studies' detail and then each natural product was discussed separately.

2 | RESULTS

A summary of natural product effects on CP toxicity is provided in Table 1 which shows the model used, dose, duration, and route of administration related to natural agents and CP, and findings of the experiments. In the next section, we discuss the studies on various natural products for their potential protective property against reproductive toxicity induced by CP. Figure 2 shows various mechanisms of natural products against CP-induced reproductive toxicity. Although the main focus of this review was herbal natural products, the obtained papers relevant to some nutraceuticals, nutritional, and natural supplements were separately considered in this study because of the importance of some natural supplements (Table 2). However, it does not cover all of the micronutrients in this subject.

3 | HERBAL EXTRACTS AND PHYTOCHEMICALS

3.1 | *Acacia hydasypica* R. Parker

Acacia hydasypica from the Leguminosae family is shown with several pharmacological activities such as antioxidant, anticancer, and anti-inflammatory effects due to its various bioactive compounds.^[86] Afsar et al., have reported the protective effect of *A. hydasypica* against CP-induced toxicities in liver, heart, kidney, and lung in rodent models, suggesting its polyphenol content as a potential preventive agent in CP-induced injuries.^[87] In a study *A. hydasypica*, ethyl acetate extract was examined against CP-induced oxidative stress, DNA injury, and testicular abnormalities in rats. *A. hydasypica* extract by the dose of 400 mg/kg/day as pretreatment and posttreatment of a single dose of CP, reversed all of the CP-induced changes through elevation of serum concentration of testosterone, follicle-stimulating hormone (FSH) and luteinizing hormone (LH), and decreased DNA damage and histopathological abnormalities in testes. *A. hydasypica* extract pretreatment was more significantly effective than the posttreatment in maintaining the levels of serum hormones which indicates its preventive potential. It also showed a decrease in oxidative stress markers such as NO, H₂O₂, and thiobarbituric acid reactive substances (TBARS) in testicular tissues. Moreover, it increased the protein content of antioxidant enzymes in testes tissue, including peroxidase (POD), superoxide dismutase (SOD), catalase (CAT), and quinone reductase (QR). Also, biochemical measurement of tissue antioxidant markers in rats showed that there was an increase in glutathione (GSH), the enzymatic activity of glutathione reductase (GR), glutathione S-transferase (GST), gamma glutamyl transpeptidase (γ-GT), and GPx by *A. hydasypica* treatment.^[17]

3.2 | *Alchemilla vulgaris* L.

A. vulgaris L. (Lady's mantle), a member of the Rosaceae family, grows in moist lawns and rangelands throughout Asia, America, and Europe. *A. vulgaris* is effective on inflammation, wounds, diarrhea, and menorrhagia.^[88] Notable pharmacological properties of *A. vulgaris* have been associated with large amounts of its phenolic acids, tannins, flavonoids, and other phenolic compounds.^[89] In a 2020 study, chromatographic analysis of the precise phenolic profile of aerial parts and roots extracts of *A. vulgaris* L. (AVA and AVR, respectively) was done. The analysis revealed the presence of more than 20 various phenolic compounds include catechin, catechin gallate, and ellagic acid as main ingredients in both extracts. AVA and AVR were administered orally in the doses of 50, 100, and 200 mg/kg to Wistar rats for 10 days and cisplatin was given on the fifth day. *A. vulgaris* extracts increased serum testosterone level and SOD, CAT, and GSH levels in testicular tissues; moreover, they reduced the levels of serum parameters of injury of testicular and parameters of oxidative stress caused by cisplatin.^[18]

3.3 | Arjunolic acid

Arjunolic acid is an active component of *Terminalia arjuna* and some other herbs which have triterpenoid structure. This substance possesses potent antioxidant and free radical scavenging activity. The effect of arjunolic acid on CP-induced testicular toxicity was assessed in Sprague Dawley rats. Testicular toxicity was induced by single-dose intraperitoneal injection of CP to rats at the dose of 7 mg/kg and they were treated with oral arjunolic acid at the dose of 20 mg/kg/day, for 10 days. As a result, testes' weight and plasma testosterone concentration elevated. Biochemical measures also supported the protective effect of arjunolic acid against CP toxicity through the result of decreasing malondialdehyde and increasing in GSH levels in testes. Gene expression of proteins related to oxidative damage in testicular tissue was measured. Cotreatment of arjunolic acid with CP positively affected oxidative stress measures and caused a decrease in testicular inducible nitric oxide synthase (iNOS), tumor necrosis factor-α (TNF-α), and p38-MAPK (mitogen-activated protein kinase). Pathological changes in testicular tissue were also improved by Arjunolic acid administration.^[19]

3.4 | β-Glucan

β-Glucans are polysaccharide compounds derived from yeast, mushrooms, and cereal species^[90] and they have immunomodulatory and antioxidant properties.^[91] Kaya et al. revealed some positive effects of β-glucan against cisplatin-induced oxidative, spermatological, and histological damages in rats. It has been administered orally for 14 days at the dose of 50 mg/kg/day in combination with a 7 mg/kg i.p. single dose of cisplatin. β-Glucan induced a significant increase

TABLE 1 Herbal extracts and phytochemicals investigated for their protective activity against CP-induced reproductive toxicity

Natural agents	Dose, duration, and route of administration	Animal Species	CP dose, duration, and route of administration	Key findings	Reference
<i>Acacia hydaspica</i> R. Parker (ethyl acetate extract)	Post treated: 400 mg/kg/day, for 6 days, orally (PO) Pretreated: 400 mg/kg/day, for 21 days, orally (PO)	Sprague Dawley rats	7.5 mg/kg, single dose, i.p.	↑ Testosterone, FSH, and LH in serum ↑ Protein content and testicular tissue antioxidant enzymes (POD, SOD, CAT, QR) ↓ NO, H₂O₂, TBARS in testicular tissues ↑ GSH and the activity level of GR, GST, γ-GT and GPx in testicular tissues ↓ Histopathological alterations ↓ DNA damage in testicular tissues as compared to posttreatment	[17]
<i>Alchemilla vulgaris</i> L. (methanol extracts of aerial parts and roots)	50, 100, and 200 mg/kg/day for 10 days, orally (PO)	Wistar rats	7.5 mg/kg, single dose, i.p.	↑ Serum testosterone level ↑ SOD, CAT, and GSH levels in testicular tissues ↓ TBARS level in testicular tissues ↓ Degeneration and atrophy of seminiferous tubules ↓ Degeneration and intraluminal desquamation of Sertoli cells ↓ Depletion of Leydig cells	[18]
Arjunolic acid	20 mg/kg/day, for 10 days, orally (PO)	Sprague Dawley rats	7 mg/kg, single dose, i.p.	↑ Testes weights and plasma testosterone levels ↓ MDA concentration and ↑ GSH contents in testicular tissues ↓ Testicular expression of iNOS, TNF-α, and p38-MAPK - Corrected the changes in histopathological parameters	[19]
β-glucan	50 mg/kg/day for 14 days, orally (PO)	Sprague Dawley rats	7 mg/kg, single dose, i.p.	↓ TBARS level in testis tissue ↑ SOD, CAT, GPx enzyme activities and GSH levels - No significant effect on testicular, epididymis, seminal vesicle, and prostate weight - No significant effect on sperm motility, epididymal sperm concentrations, and abnormal sperm rates ↓ Histopathological damage and arrested cells ↓ Intense vascular congestion, germinative epithelium spillage, edema and vacuolisation in the interstitial area and arrested spermatocytes in the testis tissue	[20]

TABLE 1 (Continued)

Natural agents	Dose, duration, and route of administration	Animal Species	CP dose, duration, and route of administration	Key findings	Reference
Caffeine	0.1, 0.5, 1%, for 15 days, orally (PO)	Balb/c mice	5.5 mg/kg, single dose, i.p.	↑ Sperm motility (0.5, 1%) ↓ Histopathological alterations ↓ The number of apoptotic cells in testes	[21]
Carvacrol	75 mg/kg/day, for 14 days, orally (PO)	Sprague Dawley rats	10 mg/kg, single dose, i.p.	↑ Sperm count and sperm motility ↓ Sperm abnormality ↑ Enzymatic and nonenzymatic endogenous antioxidants (GSH-Px, CAT, SOD, and GSH) in testicular tissues ↓ Lipid peroxidation in testicular tissues ↓ Histopathological alterations and ↑ degeneration in testicular tissues ↓ Apoptosis in testes	[22]
<i>Cochlospermum planchonii</i> (rhizome extract)	500 mg/kg/day for 14 days, orally (PO)	Sprague Dawley rats	8 mg/kg, single dose, i.p.	↑ Body weight and testicular weight ↑ Epididymis, seminal vesicle, and prostate weights ↑ Sperm motility, count, and viability ↓ Sperm abnormalities ↑ Serum testosterone, LH, and FSH ↑ Germinal epithelium height ↓ Interstitial edema ↑ Fertility parameters ↑ Antioxidant parameters (SOD, CAT, GPx, and GSH) ↓ MDA level in testicular tissue	[23]
<i>Cordyceps cicadae</i> (Nucleosides-rich extract)	50, 100, and 400 mg/kg/day for 7 days, orally	Wistar rats	10 mg/kg, single dose, i.p.	↑ Body and testes weights ↑ Testosterone and FSH levels in serum ↑ Sperm count, viability and motility ↓ MDA level in the testes ↑ Antioxidant enzymes of SOD, GSH-Px, CAT, and TAC ↓ TNF-α and IL-1β level in the testes tissues ↓ Histopathological alterations	[24]

(Continues)

TABLE 1 (Continued)

Natural agents	Dose, duration, and route of administration	Animal Species	CP dose, duration, and route of administration	Key findings	Reference
Curcumin	50 mg/kg/day, for four weeks, orally (PO)	Albino rats	4 mg/kg, twice a week for four weeks, i.p.	↓ Destruction of Leydig and Sertoli cells ↑ Number of sperm in the seminiferous tubules	[25]
				- Restored testosterone, LH, and FSH serum levels ↑ Testes and epididymus weights ↑ Sperm count and sperm motility ↓ Sperm abnormality ↓ MDA, NO, and total protein in blood ↑ GSH and TAC in blood	
	200 mg/kg/day, for 10 days, orally (PO)	Wistar albino rats	7 mg/kg, single dose, i.p.	↑ Testicular weight and MSTD ↑ Plasma testosterone level ↑ Germ cell layer number ↑ GSH and GSH-Px in testicular tissue ↓ MDA and ↓ NO in testicular tissue ↓ Histopathologic changes	[26]
Curcumin and Vitamin C	Curcumin: 20 mg/kg/day, for 60 days, orally (PO) Vitamin C: 100 mg/kg/day, for 60 days, orally (PO)	Albino rats	0.4 mg/kg/day, for 60 days, i.p.	- No effect on the expression of iNOS, p38-MAPK, and p65-NF-κB ↓ Sperm shape abnormalities ↓ Deformed sperm in head region and ↓ tail abnormalities	[27]
Curcumin, amifostine, and caffeic acid phenethyl ester	Amifostine: 400 mg/kg/day for 7 days, i.p. Curcumin: 100 mg/kg/day for 7 days, orally caffeic acid phenethyl ester: 10 μmol/kg/day for 7 days, i.p.	Sprague Dawley rats	5 mg/kg, single dose, i.p.	↑ Germinal epithelial cells and Leydig cells ↑ Germinal epithelium thickness ↑ The number of spermatogonia, spermatids and spermatozoa ↓ Degeneration in the seminiferous tubules ↓ Inter-epithelial cell vacuolization of the seminiferous tubules ↓ Hyalinization of the interstitial space and interstitial edema ↓ NF-κB/p65, caspase-3, and 8-OHdG in the germinal epithelium of the seminiferous tubules ↓ Apoptotic spermatogonia, spermatocytes, and spermatids - No effect on body weight	[16]

TABLE 1 (Continued)

Natural agents	Dose, duration, and route of administration	Animal Species	CP dose, duration, and route of administration	Key findings	Reference
Curcumin and vitamin E	Curcumin: 200 mg/kg/day, i.p., for 5 days at the same. Vitamin E: single dose 50 mg/kg i.p.	Wistar albino male adult rats	16 mg/kg, single dose, i.p.	↓ Histological damage especially in combination therapy ↓ Bax, Caspase-3	[28]
Date palm pollen (DPP) and <i>Panax ginseng</i> C.A. Mey.	DPP: 200 and 400 mg/kg/day, for 8 weeks, orally (PO) <i>Panax ginseng</i> : 200 and 400 mg/kg/day, for 8 weeks, orally (PO)	Wistar rats	3 mg/kg, three times per week for 35 days, i.p.	↑ Blood testosterone level (400 mg/kg) ↓ Testicular MDA and ↑ testicular GSH activity (400 mg/kg) ↑ Sperm count, motility, and viability (400 mg/kg) ↓ Percentage of abnormal sperms (400 mg/kg) ↑ Body and genital organs weight (400 mg/kg) ↑ Fertility parameters (fertility success and index) ↓ Histopathological alterations	[29]
<i>Diospyros lotus</i> L. (fruit extract)	1000 mg/kg/day, for 10 days, orally (PO)	Sprague Dawley rats	10 mg/kg, single dose, i.p.	↑ Sperm concentration and sperm motility ↓ Rate of abnormal sperms ↑ SOD and GSH in testicular tissues ↓ TBARS level in testicular tissues No effect on CAT and GPx activity No effect on Serum testosterone level ↓ Density of immunohistochemically Caspase-3-stained cells ↓ Histopathological alterations ↑ Weights of testes ↑ MSTD and GECT	[30]
Ellagic acid	50 mg/kg/day for 6 weeks, orally (PO)	Laca mice	5 mg/kg, once weekly for 4 weeks, i.p.	↑ GSH activity in testicular tissues ↓ LPO and ROS levels in testicular tissue ↑ Sperm motility and count ↓ Histopathological alterations ↓ Atrophic tubules ↑ Germinal epithelium height	[31]

(Continues)

TABLE 1 (Continued)

Natural agents	Dose, duration, and route of administration	Animal Species	CP dose, duration, and route of administration	Key findings	Reference
	10 mg/kg/day, for 10 days, orally (PO)	Sprague Dawley rats	7 mg/kg, single dose, i.p.	↑ Sperm count and sperm motility ↓ Sperm abnormality ↑ Testicular weights and dimensions and other reproductive organ weights ↓ MDA level in plasma, sperm, and testicular tissue - No effect on glutathione level ↑ GSH-Px activity in testicular tissue ↑ Catalase activity in plasma, sperm, and testicular tissue ↓ Histopathological alterations	[10]
Epigallocatechin-3-gallate	40 and 80 mg/kg/day, for 5 days, i.p.	Sprague Dawley rats	10 mg/kg, single dose, i.p.	↑ Serum testosterone and testicular TAS ↓ MDA, NO, IL-6, and IL-1β in testicular tissue ↓ Testicular cytosolic cytochrome C, Bax/Bcl-2 ratio, and caspase-3 activity ↓ Testicular injury score and ↑ spermatogenesis ↓ Testicular TNF-α expression	[32]
Eugenol	100 mg/kg/day for 5 days, i.p.	Sprague Dawley albino rats	15 mg/kg, single dose, i.p.	↓ MDA concentration in testicular tissues ↑ GSH levels and GPx, SOD, CAT activities in testicular ↑ Sperm motility ↓ Abnormal sperm rate ↑ Tubules wall thickness ↓ Degeneration and necrosis in the seminiferous tubules ↓ Caspase-3 expression	[33]
<i>Eugenia jambolana</i> extract (EJE)	25 mg/kg alternate day for 7 days, i.m.	Holtzman rats	5 mg/kg, single dose, i.p.	↑ Testosterone and LH in serum ↑ mRNA expression levels of 3β-HSD ↑ Total glutathione level and TAC in testicular tissues ↓ Testicular MDA no effect on HO-1 levels	[34]

TABLE 1 (Continued)

Natural agents	Dose, duration, and route of administration	Animal Species	CP dose, duration, and route of administration	Key findings	Reference
Grape seed extract	100, 200, and 400 mg/kg/day for 15 days, orally (PO)	Wistar rats	10 mg/kg, single dose, i.p.	↑ Body weight and testes weight ↑ Food utilization rate and ↓ organ coefficients ↓ Abnormalities of seminiferous tubule structure and epithelial cells in Stages IV–VI of cycle ↑ The seminiferous tubule structure and the number of epithelial cells ↓ Testicular interstitial hyperemia and vacuoles ↑ Percentage of seminiferous tubules and seminiferous epithelium ↑ Diameter of seminiferous tubule ↑ Percentage of testicular interstitium ↓ Percentage of tubule lumen and diameter of tubule lumen in Stages IV–VI and XI of seminiferous epithelium cycle ↑ Activities of SOD, CAT and GSH and levels of T-AOC ↓ OH levels ↓ Levels of OH and LPO in the testicular tissue and 8-OHdG in serum ↑ mRNA levels of MT-1 and ↓ HO-1 in testes ↓ NO level and activities of iNOS and TNOS ↑ Serum level of testosterone ↑ mRNA expression levels of Lhcgr, Scarb1, Star, Cyp11a1, Cyp17a1, Cyp19a1, Hsd3b1, and Hsd17b ↑ Key enzymes of testosterone synthesis including STAR, CYP11a1, and HSD17B	[39]
	200, and 400 mg/kg/day; PO; for 15 days, starting 10 days	Sprague Dawley rats	7 mg/kg single dose, i.p.	↑ Testicular and epididymal weight, epididymal sperm count, motility, ↑ Activities of GSH-Px, SOD, GSH level, ↓ MDA (400 mg/kg)	[40]
	250 mg/kg/day, i.p; for 6 days	Rabbits	5 mg/kg, single dose, i.p	↓ MDA level, ↑ CAT and GSH-Px and GSH levels	[41]
<i>Ichnocarpus frutescens</i> (L.) W.T. Aiton (aqueous root extract) and <i>Cyperus</i>	100, 200, and 400 mg/kg, single dose, orally (PO)	Wistar albino rats	10 mg/kg, single dose, orally (PO)	↑ The number of spermatids and spermatozoa ↑ Sperm density and motility	[42]

TABLE 1 (Continued)

Natural agents	Dose, duration, and route of administration	Animal Species	CP dose, duration, and route of administration	Key findings	Reference
<i>rotundus</i> L. (aqueous rhizome extract)				↑ Blood levels of reproductive hormones (testosterone, FSH, and LH) ↓ MDA concentration in testicular tissues in dose-dependent manner ↑ CAT and GSH level in testicular tissues	
<i>Kigelia africana</i> (Lam.) Benth. (fruit extract)	100 and 500 mg/Kg/day, for 56 days (cotreatment group) or 28 days (pretreatment and posttreatment group), orally (PO)	Sprague Dawley rats	10 mg/kg, single dose, i.p.	↑ Body weight and testicular weight/volume ↑ Sperm count and sperm motility ↓ MDA level in testicular tissue ↑ CAT and GSH activity in testicular tissue ↑ Blood levels of reproductive hormones (testosterone, FSH, and LH) ↓ Histopathological alterations	[43]
	100 and 500 mg/kg/day, for 28 days (cotreatment group) or 14 days (pretreatment and posttreatment group), orally (PO)	Sprague Dawley rats	10 mg/kg, single dose, i.p.	↑ Body weight and testicular weight/volume ↑ Sperm count and sperm motility ↑ Diameter and cross-sectional areas of seminiferous tubules ↑ Serum testosterone, ↓ LH, and FSH ↓ MDA activity in testicular tissue ↑ CAT and GSH levels in testicular tissue ↓ Histopathological changes in testicular tissue	[44]
<i>Launaea taraxacifolia</i> (aqueous leave extract)	100 and 400 mg/kg/day for 21 days, orally (PO)	Wistar albino rats	10 mg/kg, single dose, i.p.	↑ Testicular weight ↓ Histopathological alterations ↑ Sperm motility and count ↓ Abnormal sperm cells ↑ CAT and SOD activity and glutathione levels in testicular tissues ↓ MDA level in testicular tissues	[45]
Lycopene	4 mg/kg/day, for 10 days (pretreatment group) or 5 days (posttreatment group), orally (PO)	Sprague Dawley rats	7 mg/kg, single dose, i.p.	↑ Sperm concentration ↑ Sperm motility (posttreatment group) No effect on MDA level ↑ GSH levels in testicular tissues (posttreatment group) No effect on GSH-Px activity and Testosterone level	[46]

(Continues)

TABLE 1 (Continued)

Natural agents	Dose, duration, and route of administration	Animal Species	CP dose, duration, and route of administration	Key findings	Reference
Lycopene and selenium	Lycopene: 4 mg/kg/day, for 15 days, orally (PO), Selenium: 0.1 mg/kg/day, for 15 days, orally (PO)	Wistar rats	10 mg/kg, single dose, i.p.	↓ Sperm abnormality (posttreatment group) ↑ Diameter size of seminiferous tubules (posttreatment group) * Effect of giving lycopene after CP was superior to the giving it before CP ↑ Testes weights and dimensions ↑ Epididymides and accessory gland weights ↑ Sperm count and motility (selenium had no effect on sperm motility) ↓ Abnormal sperm rate ↓ MDA level in testicular tissue ↑ SOD, CAT, and GSH-Px activity in testicular tissues ↑ GSH level in testicular tissues ↑ Blood levels of testosterone	[47]
<i>Maclura pomifera</i> (Raf.) C.K.Schneid. (aqueous fruit extract)	500 mg/kg/day for 5 days before and after the CIS injection, orally (PO)	Sprague Dawley rats	7 mg/kg, single dose, i.p.	↓ Dead sperm rates No effect on sperm motility, sperm abnormalities, and sperm cell densities	[48]
<i>Melissa officinalis</i> L. and Vitamin E	M. officinalis: 1000 mg/kg/day, for 7 days, orally (PO) Vitamin E: 100 mg/kg/day for 7 days, orally (PO)	Wistar rats	10 mg/kg, single dose, i.p.	↑ Serum testosterone, ↑ LH, ↑ FSH, ↑ weight of testes ↑ sperm motility, count and vitality; ↓ Sperm cell abnormalities	[49]
MOTILIPERM (<i>Morinda officinalis</i> F.C. How., <i>Cuscuta chinensis</i> Lam. and <i>Allium cepa</i> L.)	100 and 200 mg/kg/day, for 28 days, orally (PO)	Sprague Dawley rats	10 mg/kg, single dose, i.v.	↑ Body weight (both doses), ↑ Prostate weight (200 mg/kg) No effect on testes weight ↑ Epididymis weight (100 mg/kg), ↓ Sperm apoptosis No effect on sperm count and sperm motility ↓ MDA concentration in testicular tissues ↑ Blood level of testosterone (100 mg/kg) No effect on spermatogenic cell density ↑ Johnsen's score in seminiferous tubules (200 mg/kg)	[50]

TABLE 1 (Continued)

Natural agents	Dose, duration, and route of administration	Animal Species	CP dose, duration, and route of administration	Key findings	Reference
Naringenin	50 mg/kg/day for 10 days, orally (PO)	Sprague Dawley rats	10 mg/kg, single dose, i.p.	No effect on StAR protein ↓ GRP-78 protein, p-IRE1 and p-JNK in testicular tissues ↑ Serum testosterone and testicular TAS ↓ MDA level in testicular tissue ↓ TNF- α /IL-10 ratio in testes ↑ Bcl-2/Bax ratio ↓ Caspase-3 expression ↑ P-glycoprotein in testicular tissue ↓ Histopathological alterations and testicular injury score ↑ Spermatogenesis ↓ Testicular iNOS activity	[51]
<i>Nigella sativa</i> L. (crude oil)	0.5 ml/kg/day, for 21 days, i.p.	Albino rats	1 mg/kg, daily for 21 days, i.p.	↓ MDA concentration in testicular tissues ↓ Degeneration, necrosis, and parenchymal atrophy in the seminiferous tubules	[52]
Olive leaf extract	300 mg/kg/day for 5 days, orally (PO)	Wistar rats	5 mg/kg, single dose, i.p.	↑ Body and testicular weight ↑ Serum testosterone, LH, and FSH ↓ Testicular MDA and nitrite/nitrate levels ↑ Testicular GSH levels ↑ Antioxidant enzymes (SOD, CAT, GSH-Px, and GSH-R) ↑ mRNA expression levels of Nrf2 and HO-1 ↓ IL-1 β and TNF- α levels ↓ Degeneration, necrosis, and parenchymal atrophy in the seminiferous tubules ↓ Bax and ↑ Bcl-2 mRNA expression levels ↑ PCNA-positive cells	[53]
Pomegranate seed extract	250 mg/kg/day for 6 days before and after the CIS injection, orally (PO)	New Zealand white rabbits	5 mg/kg, single dose, i.p.	↓ Testicular MDA ↑ CAT and GPx activity and glutathione levels in testicular tissues	[54]

(Continues)

TABLE 1 (Continued)

Natural agents	Dose, duration, and route of administration	Animal Species	CP dose, duration, and route of administration	Key findings	Reference
<i>Prosopis cineraria</i> (L.) Druce (unripe fruit pod)	400 and 800 mg/kg/day for 28 days, orally (PO)	Wistar rats	1 mg/kg/day, for 15 days, i.p.	↑ Body weight and testis weight ↑ Sperm motility, sperm count, sperm viability and improved morphological features. ↑ Testosterone, FSH, and LH levels in serum ↑ LPO, CAT, GSH, GPx, and SOD in testicular tissues.	[55]
Quercetin	50 mg/kg/day, for 10 days, orally (PO)	Wistar albino rats	7 mg/kg, single dose, i.p.	- No effect on serum TAC and TOS - No effect on testicular levels of MDA, CAT, SOD and XO ↓ Histopathological alterations (Quercetin + CP)	[56]
Resveratrol(RES)	50, 100 mg/kg/day, for 14 days, orally (PO)	Swiss albino mice	4, 8 mg/kg, single dose, i.p.	↓ DNA strand-breaks ↓ MNPCE (RES: 100 mg/kg + cDDP 4, 8 mg/kg) ↓ Mitotic activity (% PCE) in bone marrow (RES: 100 mg/kg + cDDP 8 mg/kg) ↓ Spermatogonial chromosomal aberrations ↓ spermatocyte chromosomal aberrations ↑ Percentage of live cells in bone marrow (RES: 100 mg/kg + cDDP 8 mg/kg) ↓ Percentage of early apoptotic cells in bone marrow (RES: 100 mg/kg + cDDP 8 mg/kg) ↓ Percentage of late apoptotic/necrotic cells in bone marrow (RES: 100 mg/kg + cDDP 8 mg/kg) - No effect on percentage of damaged cells in bone marrow ↓ Percentage of bone-marrow DNA fragmentation (RES: 100 mg/kg + cDDP 8 mg/kg) ↓ Percentage of caspase-3 activity in bone marrow (RES: 100 mg/kg + cDDP 8 mg/kg) ↓ ROS production in bone marrow (RES: 100 mg/kg + cDDP 8 mg/kg) ↓ MDA in bone marrow (RES: 100 mg/kg + cDDP 8 mg/kg) ↑ GSH and GSSG in bone marrow (RES: 100 mg/kg + cDDP 8 mg/kg) ↓ Level of urinary 8-OHdG (RES: 100 mg/kg + cDDP 8 mg/kg)	[57]

TABLE 1 (Continued)

Natural agents	Dose, duration, and route of administration	Animal Species	CP dose, duration, and route of administration	Key findings	Reference
	20 mg/kg/day, for 45 days, i.p.	Wistar rats	3 mg/kg, on first 3 alternate days, i.p.	↑ Sizes and volumes of testes and other reproductive organs	[12]
				↑ Sperm count and sperm motility	
				↓ Sperm abnormality	
				↑ HSD (3β and 17β) activity in testicular tissues	
				↑ Serum testosterone levels	
				↓ MDA and H ₂ O ₂ levels in testicular and epididymal tissues	
				↑ CAT and SOD activity in testicular and epididymal tissues	
				↓ Histopathological alterations	
				↑ MSTD and GECT	
				↓ Activities of AST, ALT, ALP, and LDH in serum	
				↓ Testicular weight	
				↑ Sperm motility and sperm concentration	
	1 mg/kg/day, for 28 days, orally (PO)	Albino mice	7.5 mg/kg/week, for 4 weeks, i.p.	↑ GSH, Redox Ratio (GSH/GSSG), GPx, CAT, and SOD in testicular tissues	[58]
				↓ GSSG and GST in testicular tissues	
				↓ MDA and NO level in testicular tissues	
				↓ DNA fragmentation in testes	
				↓ Necrotic sperms in testicular tissues	
				↓ Histopathological alterations	
				↑ Germinal epithelium height	
				↓ Seminiferous Tubule lumen diameter	
				↑ Testosterone, FSH, and LH in serum	
				↑ Final body and testis weights	
				↑ Sperm count and motility	
				↓ Sperm morphological abnormalities	
	20 mg/kg/day, for 45 days, i.p.	Wistar rats	3 mg/kg, for 3 days, i.p.	↑ SOD, CAT, GPx enzyme activities, and GSH levels	[59]

(Continues)

TABLE 1 (Continued)

Natural agents	Dose, duration, and route of administration	Animal Species	CP dose, duration, and route of administration	Key findings	Reference
				↓ Expression of cleaved caspase-3, calpain-1, cleaved caspase-12, p-ERK1/2, and p-SAPK/p-JNK ↓ Testicular apoptosis ↓ p53 and Bax and ↑ Bcl-2 protein levels in the testis ↑ Protein levels of p-Akt and p-Bad	
<i>Rhodiola rosea</i> L.	0.0125–2.5 mg/L	Mice TM4 Sertoli cell line	0.0147 g/L	↑ Proliferation of the TM4 cells improved the TM4 cells morphological changes ↓ MDA content (0.1 mg/L) ↑ SOD and GSH activity (0.1 mg/L)	[60]
Rutin	150 mg/kg/day, for 14 days, orally (PO)	Sprague Dawley rats	10 mg/kg, single dose, i.p.	↑ Sperm motility ↓ Abnormal sperm rate ↑ GSH-Px, CAT, and SOD activity ↑ GSH and ↓ MDA levels	[61]
	75 mg/kg/day for 13 days, orally (PO)	Sprague Dawley rats	7 mg/kg, single dose, i.p.	↑ Testicular daily sperm production ↑ Head length and ↓ tail length and tail moment ↑ Diameters of seminiferous tubules and the germinal epithelial height ↑ <i>Tunica albuginea</i> height ↓ Seminiferous tubular lumen diameter and interstitial space ↑ Germ cell number (↑ spermatogonia, primary spermatocytes, secondary spermatocytes, and spermatids) ↑ CAT, SOD, POD, and glutathione level in testicular tissues ↓ TBARS level in testicular tissues No effect on plasma testosterone level ↑ Intra-testicular testosterone concentration	[62]
<i>Satureja hortensis</i> L. (summer savory) aerial parts methanolic extract	50, 100, and 200 mg/kg/day for 10 days, orally (PO)	Wistar rats	7.5 mg/kg, single dose, i.p.	↑ Serum level of testosterone ↑ Levels of SOD, CAT, GSH in testicular tissues ↓ TBARS level in testicular tissues	[63]

TABLE 1 (Continued)

Natural agents	Dose, duration, and route of administration	Animal Species	CP dose, duration, and route of administration	Key findings	Reference
Speman® (<i>Withania somnifera</i> , <i>Asteracantha longifolia</i> , <i>Lactuca scariola</i> , <i>Mucuna pruriens</i> , <i>Parmelia perlata</i> , <i>Argyrea speciosa</i> , <i>Tribulus terrestris</i> , <i>Leptadenia reticulata</i> , and <i>Suvarna vang</i>)	80 mg/kg/day, for 35 days, orally (PO)	Swiss albino mice	3 mg/kg, for 3 alternate days, i.p.	↑ Bcl-2/Bax ratio in testes tissues ↓ Degeneration of seminiferous tubules and Sertoli cells ↑ Testicular and epididymal weight ↑ Sperm count, sperm motility, and viable sperms ↓ Sperm abnormality ↑ HSD (3β and 17β) activity in testicular tissues ↑ Serum testosterone levels	[64]
<i>Tribulus terrestris</i> L. (Hydroalcoholic extract)	100, 300 and 500 mg/kg/day, for 4 days, i.p.	Balb/c mice	5.5 mg/kg, single dose, i.p.	↑ The weights of body and testes, and seminiferous tubules diameter (300 mg/kg/day) ↓ Apoptotic cells of seminiferous tubules (300, 500 mg/kg/day)	[65]
<i>Trigonella foenum-graecum</i> Linn. (Fenugreek: seed extract)	50, 100, and 200 mg/kg/day, for 5 days, orally (PO)	Wistar rats	10 mg/kg, single dose, i.p.	↑ Weights of testes and epididymides in dose-dependent manner ↑ Sperm count and motility in dose-dependent manner ↓ Histopathological changes in testicular tissue ↓ Germ cell apoptosis ↓ Testicular MDA levels and SOD and MPO activities ↑ CAT activity ↓ Expression of iNOS and NF-κB in testes	[66]
<i>Zataria multiflora</i> Boiss.	200 mg/kg/day for 7 days, orally (PO)	Balb/c mice	10 mg/kg, single dose, i.p.	↓ MDA and PC levels, ↑ GSH level in testicular tissues No effect on testicular weight ↑ Sperm count, motility, and viability rate ↓ Sperm abnormality ↑ Serum testosterone level ↓ Histopathological alterations ↓ Necrosis and degeneration of seminiferous tubules, and interstitial edema ↓ Destruction in Leydig cell in seminiferous tubules ↑ Spermatogenesis	[67]

(Continues)

TABLE 1 (Continued)

Natural agents	Dose, duration, and route of administration	Animal Species	CP dose, duration, and route of administration	Key findings	Reference
<i>Zingiber officinale</i> Roscoe.	310 mg/Kg/day, for 26 days, orally (PO)	Albino rats	12 mg/kg, single dose, i.p.	↑ Johnsen's scores in testicular tissue ↑ Seminiferous tubular diameter and epithelial thickness ↓ Caspase-3 immunoreactivity ↑ Body and testicular weight ↑ Diameter of seminiferous tubules ↑ Percentage of normal seminiferous tubules ↓ Histopathological alterations ↑ Body and testicular weight ↑ CAT, SOD, GSH and GPx level in testicular tissues ↓ Testicular level of MDA ↑ Serum testosterone, LH, and FSH ↓ Testicular IL-1β, TNF-α, IL-6, NF-κB and NO levels, and iNOS activity ↓ Histopathological alterations ↑ Testicular weight ↑ Testosterone level in serum ↓ Histopathological alterations ↓ Structural changes in both Sertoli cells and early spermatids ↑ Germinal thickness of the seminiferous tubules	[66]
	5 ml/kg/day for 5 days, orally (PO)	Wistar rats	10 mg/kg, single dose, i.p.	↑ Body and testicular weight ↑ CAT, SOD, GSH and GPx level in testicular tissues ↓ Testicular level of MDA ↑ Serum testosterone, LH, and FSH ↓ Testicular IL-1β, TNF-α, IL-6, NF-κB and NO levels, and iNOS activity ↓ Histopathological alterations ↑ Testicular weight ↑ Testosterone level in serum ↓ Histopathological alterations ↓ Structural changes in both Sertoli cells and early spermatids ↑ Germinal thickness of the seminiferous tubules	[69]
	200 mg/kg/day for 30 days, orally (PO)	Wistar albino rats	7 mg/kg, single dose, i.p.	↑ Testicular weight ↑ Testosterone level in serum ↓ Histopathological alterations ↓ Structural changes in both Sertoli cells and early spermatids ↑ Germinal thickness of the seminiferous tubules	[70]
<i>Zingiber officinale</i> Roscoe. (rhizome extract) and <i>Hibiscus sabdariffa</i> L. (Dried flowers extract)	Roselle: 1 g/kg/day, for 26 days, orally (PO) Ginger: 1 g/kg/day, for 26 days, orally (PO)	Wistar albino rats	10 mg/kg, single dose, i.p.	↓ Apoptotic cells of seminiferous tubules ↓ Percentage of apoptosis in sperms ↓ Protein expression of p53 in the seminiferous tubules ↓ Histopathological alterations ↓ MDA level in testicular tissue - No effect on the weights of testes and epididymides ↑ Sperm count and motility ↓ Sperm abnormality^a	[71]
	Roselle: 1 g/kg/day, for 26 days, orally (PO) Ginger: 1 g/kg/day, for 26 days, orally (PO)	Wistar albino rats	10 mg/kg, single dose, i.p.	- No effect on the weights of testes and epididymides ↑ Sperm count and motility ↓ Sperm abnormality^a	[14]

TABLE 1 (Continued)

Natural agents	Dose, duration, and route of administration	Animal Species	CP dose, duration, and route of administration	Key findings	Reference
				↓ MDA concentration in testicular tissues ↑ CAT, SOD, and GSH level in testicular tissues ↓ Histopathological alterations	

Abbreviations: 8-OHdG, 8-hydroxy-2'-deoxyguanosine; AHE, acacia hydasipica ethyl acetate; AsA, ascorbic acid; AST, aspartate amino transferase; ATM, ataxia telangiectasia mutated; BAX, BCL2-associated X protein; Bcl-2, B-cell lymphoma 2; BUN, blood urea nitrogen; CAT, catalase; cDDP, displatin; CMA3, chromomycin A3 staining; COX-2, cyclooxygenase-2; CYP11A1, P450 side chain cleavage enzyme; CYP17A1, 17 α -hydroxylase; Cyp19a1, 19 α -hydroxylase; FSH, follicle stimulating hormone; GECT, germinal epithelial cell thickness; GR, glutathione reductase; GRP-78, Glucose-regulated protein-78; GSH, glutathione; GSH-Px, glutathione peroxidase; GSSG, oxidized glutathione; GST, glutathione S-transferase; HSD, hydroxysteroid dehydrogenase; HSD17B, 17 β -hydroxysteroid dehydrogenase; HSD3B1, 3 β -hydroxysteroid dehydrogenase; IL-1 β , interleukin-1 β ; IL-6, interleukin-6; iNOS, inducible nitric oxide synthase; LH, luteinizing hormone; Lhcgr, luteinizing hormone receptor; LPO, lipid peroxides; MAPK, mitogen-activated protein kinase; MDA, malondialdehyde; MMP, mitochondrial membrane potential; MPO, myeloperoxidase; MSTD, mean seminiferous tubule diameter; NF- κ B, nuclear factor-kappa B; NO, nitric oxide; OSI, oxidative stress index; PCNA, proliferating cell nuclear antigen; p-IRE1, phosphorylated inositol-requiring transmembrane Kkinase/endoribonuclease 1; p-JNK, phosphorylated c-jun-N-terminal kinase; POD, peroxidase; QR, quinone reductase; ROS, reactive oxygen species; Scarb1, lipoprotein-receptor; SOD, superoxide dismutase; STAR, steroidogenic acute regulatory protein; TAC, total antioxidant capacity; T-AOC, total antioxidant capacity; TAS, total antioxidant status; TBARS, thiobarbituric acid reactive substances; TNOS, total nitric oxide synthases; TOS, total oxidant status; TUNEL, transferase dUTP nickend labeling; XO, xanthine oxidase; γ -GT, gamma glutamyl transpeptidase; γ -H2AX: H2A histone family member X.

in antioxidant levels and SOD, CAT, GPx enzyme activities. Also, it induced a decrease in TBARS level in testis tissue.^[20]

3.5 | Caffeine

Oral administration of caffeine (0.1%, 0.5%, 1%, for 15 days) in CP-treated mice was shown to prevent changes in the total body and testes weights, as well as histopathological lesions. Also, caffeine improved the motility of sperms impaired by CP after 15 days. Moreover, caffeine subchronic supplementation decreased the number of apoptotic cells induced by CP suggesting its antioxidant effect in the recovery of CP-induced toxicity with special emphasis on a low dose of caffeine.^[21] Some studies reported that daily caffeine intake (>2.9 cups) may affect reproductive function in men, possibly via DNA damage in sperm. However, there is no strong potential association between caffeine use and male infertility based on the currently available evidence.^[92]

3.6 | Carvacrol

Carvacrol is a monoterpene phenolic substance that exists in various plants such as *Origanum* and *Thymus* species. Antioxidant properties of carvacrol have been demonstrated in several studies. Oral administration of carvacrol prevented DNA damage that could be induced by hydrogen peroxide in liver and testis tissues. It also prevented N-nitrosodiethylamine-induced lipoperoxidation and carcinogenesis.^[93,94] In another study, it is shown CP (10 mg/kg, single dose; intraperitoneal [i.p.]) caused gonadotoxicity in rats. The co-treatment of CP with carvacrol (75 mg/kg/day; p.o.) for 14 days, relieved spermiotoxicity parameters including sperm count, motility, and abnormalities via diminishing lipid peroxidation and apoptosis in testicular tissue. Carvacrol also reversed testicular degeneration and pathological changes mediated by CP.^[22]

3.7 | *Cochlospermum planchonii* Hook.f. ex Planch

A 2018 study evaluated the fertility-enhancing effects of *C. planchonii* rhizome extract against reproductive disorders caused by CP. Testicular toxicity was induced by the injection of CP single dose (i.p.) to rats at the dose of 8 mg/kg and they were administered 500 mg/kg/day of *C. planchonii* orally for 14 days. Administration of *C. planchonii* significantly increased body and testicular weight as well as sperm motility, count, and viability. Additionally, *C. planchonii* elevated malondialdehyde (MDA) and decreased CAT, SOD, GSH, and GPx.^[23]

3.8 | Curcumin

Curcumin is the main polyphenol constituent of turmeric (*Curcuma longa* L.). It is effective in the management of various conditions such as systemic inflammation, arthritis, pain, and metabolic syndrome. Most of

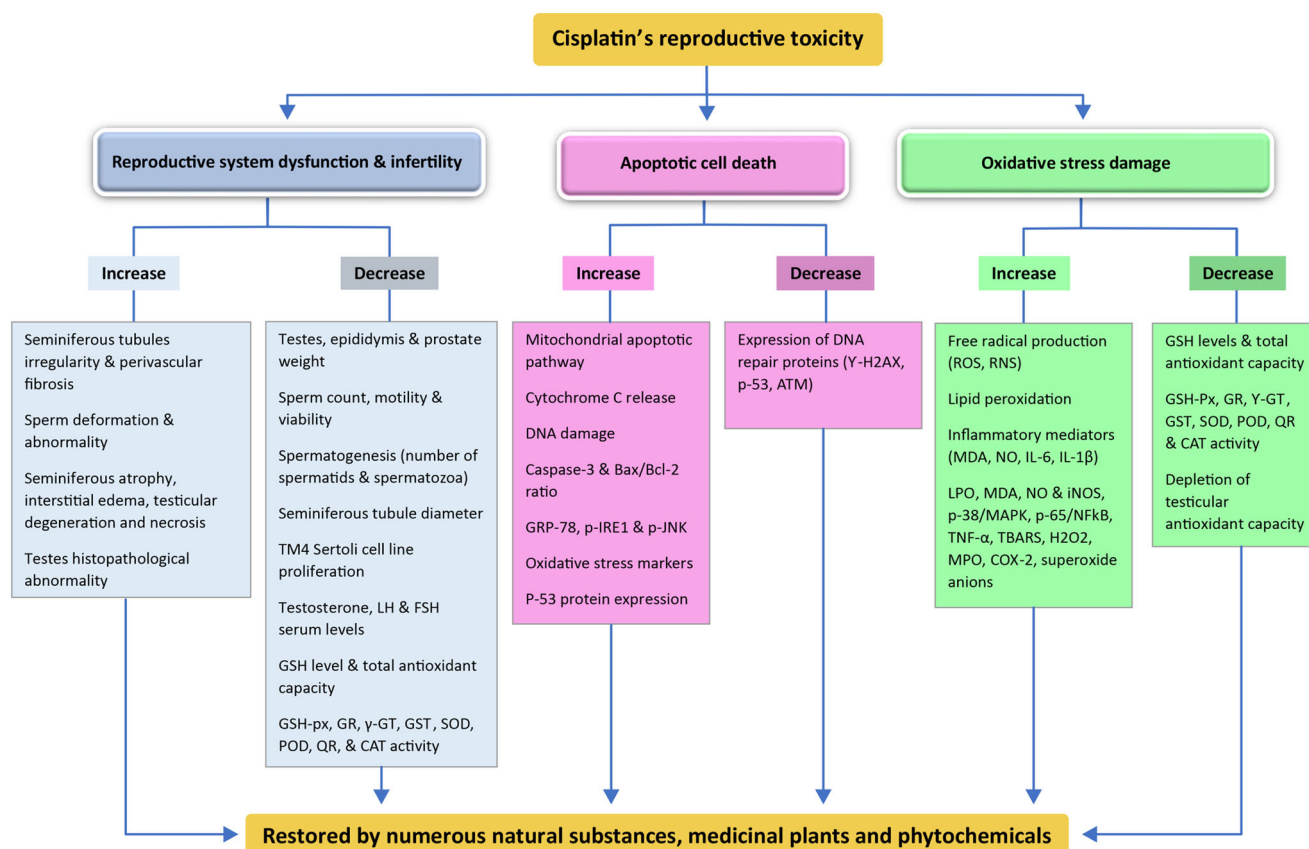


FIGURE 2 Mechanisms of natural products against cisplatin-induced reproductive toxicity

these effects are related to its antioxidant and anti-inflammatory properties which are mediated through its potent free radical scavenging activity.^[95] It has been shown that oral administration (p.o.) of curcumin with the dose of 50 mg/kg/day for 4 weeks restored testosterone, LH, and FSH serum levels in rats treated or posttreated with intraperitoneal CP (4 mg/kg, twice a week for 4 weeks) to reduce male sexual hormones. Curcumin also restored testes and epididymal weight to normal. Furthermore, curcumin significantly increased sperm count, sperm motility and GSH, and total antioxidant capacity (TAC) in blood. Moreover, it significantly decreased MDA, nitric oxide (NO), and total protein in blood in comparison with CP-treated rats.^[25] In another study on Wistar Albino rats, it has been shown that oral administration of curcumin with the dose of 200 mg/kg/day for 10 consecutive days ameliorated biochemical and histopathological damages of testes caused by a single 7 mg/kg, i.p. dose of CP. In fact, curcumin reversed CP-induced seminiferous tubules irregularity, seminiferous epithelial layers reduction, and maturation arrest and perivascular fibrosis. Curcumin also increased parameters including plasma testosterone, testicular weight, testicular GSH levels, and glutathione peroxidase (GSH-Px) activity. Moreover, curcumin administration reversed the effects of CP, which caused oxidative damage by increasing markers of oxidative stress such as MDA and NO in testicular tissue. The effect of oxidative damage of CP was manifested through influencing MAPK and NF-κB pathways. Furthermore, CP triggered overexpression of p38/MAPK,

which induced p65/NF-κB expression, and consequently, it stimulated iNOS which eventually resulted in pathologic NO excessive production.^[11] Also, CP-induced caspase-3 expression in germ cells was inhibited by treatment with curcumin, vitamin E, and their combination therapy suggesting their potential preventive effect against CP-induced testicular apoptosis via the caspase-3 and Bax pathways.^[28] CP administration (0.4 mg/kg/day; i.p.; 20 min after curcumin) for 60 consecutive days has also been shown to cause sperm deformation in rats which was ameliorated by treatment with curcumin at 20 mg/kg/day.^[27] Moreover, the protective effects of curcumin, amifostine, and caffeic acid on testicular toxicity of CP were investigated. The findings showed that they decreased the expression of NF-κB/p65, 8-OHdG, and caspase-3, especially amifostine. The mentioned pathways and biomarkers were shown to be significantly relevant to testis toxicity caused by CP. So, amifostine, curcumin and to a less amount, caffeic acid phenethyl ester might have the potential to be used as therapeutic adjuvants in testicular damage caused by CP.^[16]

3.9 | Date palm pollen (DPP) and *Panax ginseng* C. A. Mey

DPP and *P. ginseng* extracts displayed a protective effect in testicular toxicity caused by CP. Wistar rats were treated with intraperitoneal

TABLE 2 Some selected nutraceuticals, nutritional, and natural supplements

Natural agents	Dose, duration, and route of administration	Animal Species	CP dose, duration, and route of administration	Key findings	Reference
All-trans retinoic acid	7.5 mg/kg/day, for 10 days, i.p.	Wistar albino rats	7 mg/kg, single dose, i.p.	No significant effect on testicular weight ↑ Sperm count and motility ↓ Percentage of abnormal sperm No significant effect on TAS ↓ TOS and OSI in testicular tissue ↓ Arrest of spermatogenesis ↓ Testicular damage score ↑ Johnsen's scores in testicular tissue	[72]
α-tocopherol, L-ascorbic acid, Zn, and Selenium	α-tocopherol: 100 mg/kg, L-ascorbic acid: 50 mg/kg, Zn: 40 mg/l, Selenium: 100 µg/l, for one week, orally (PO)	Sprague Dawley rats	10 mg/kg, for 4 days, i.p.	↑ Testicular weights ↑ Sperm count No effect on sperm motility ↑ Seminiferous tubular diameter and seminiferous epithelial thickness ↓ MDA level in testicular tissue No effect on catalase and glutathione reductase activity ↑ GPx and SOD activity in testicular tissue ↓ p53, Bcl-2, and Bax protein levels in the testis ↓ Cell death in the testis ↓ Histopathological alterations	[73]
L-Ascorbic acid	10 mg/Kg/day, for 30 days, i.p	Balb/c mice	1, 2.5 mg/kg/day, for two separate rounds (each round was 5 days), i.p.	↓ Testes weight (CP 1 mg/kg + AsA) ↑ Testes weight (CP 2.5 mg/kg + AsA) ↓ Incidence of abnormal tubules ↑ STD (CP 1 mg/kg + AsA) ↓ SE (CP 2.5 mg/kg + AsA) ↑ Sperm count no effect on sperm motility ↓ Sperm abnormalities (CP 2.5 mg/kg + AsA)	[74]

(Continues)

TABLE 2 (Continued)

Natural agents	Dose, duration, and route of administration	Animal Species	CP dose, duration, and route of administration	Key findings	Reference
Vitamin C and L-Cysteine	Vitamin C: 100 mg/kg L-cysteine: 100 mg/kg, once a week for four weeks, i.p.	Albino rats	2 mg/kg, once a week for four weeks, i.p.	↓ LPO and superoxide anion levels in testicular tissue (L-cysteine) ↓ Total peroxides in testicular tissue and plasma (L-cysteine) ↓ Total peroxides in plasma (vitamin C) ↓ Superoxide anion levels in testicular tissue and plasma (L-cysteine) ↑ CAT activity in plasma (L-cysteine) ↑ Activity of SOD in testicular tissue ↑ CAT and GST activity in testicular tissue (vitamin C) ↑ GSH and Vit E concentration in testicular tissue and plasma ↑ NO level in testicular tissue (vitamin C) degeneration and necrosis in some seminiferous tubules ↓ Histopathological alterations and number of apoptotic cells	[75]
Vitamin E (α-tocopherol)	50 and 200 mg/kg/day for 7 weeks, orally (PO)	Sprague Dawley rats	5 mg/kg, single dose, i.p.	↑ Sperm motility and sperm morphology	[76]
Vitamin E and Zinc nanoparticles	zinc nanoparticle: 10, 25 and 50 mg/kg, orally vitamin E: 100 mg/kg, i.p., for 14 days	NMRI mice	7 mg/kg, single dose, i.p.	↓ MDA level and ↑ GSH levels ↑ SOD and CAT enzyme activities ↑ Thickness of the germinating epithelium ↓ Seminiferous tract damage and vascular congestion	[77]
L-Carnitine	250 mg/kg/day for the next 3 days, i.p.	Wistar Albino rats	5 mg/kg, single dose, i.p.	- No effect on testis weight/body weight ratio ↑ Diameter of seminiferous tubules and ↓ interstitial area ↓ Percentage of degenerated tubules ↓ Histologic abnormalities ↑ normal seminiferous tubules and normal spermatogenesis ↓ Percentage of TUNEL positive tubules ↓ TUNEL positive cells and ↓ PCNA positive cells ↑ Proliferation rate of germ cells in testis ↑ Sperm count and viability ↓ Percentage of sperm with abnormal morphology	[78]

TABLE 2 (Continued)

Natural agents	Dose, duration, and route of administration	Animal Species	CP dose, duration, and route of administration	Key findings	Reference
Melatonin	10 mg/kg/day, for 5 days (short-term treatment) and 50 days (long-term treatment), i.p.	Sprague Dawley rats	7 mg/kg, single dose, i.p.	↑ Testes, epididymis, and prostate weights (in the short-term treatment) ↑ Sperm concentration (in the short-term treatment) ↑ Sperm motility ↓ Abnormal sperm rates ↓ MDA concentration in testicular tissues ↑ GSH-Px activity and glutathione level in testicular tissues ↓ Histopathological alterations (testes degeneration, necrosis, and interstitial edema)	[13]
	20 mg/kg/day for 10 days, orally (PO)	Wistar rats	7 mg/kg, single dose, i.p.	No effect on testes weight ↑ Sperm motility and count ↓ Sperm abnormalities ↑ Serum level of testosterone ↓ Testicular level of MDA ↑ GSH levels in testicular tissues ↓ Histopathological alterations	[79]

Fucoxanthin-rich extract (FXE) from brown algae	100, 200, and 500 mg/kg/day for 5 days, orally (PO)	Syrian hamsters	7 mg/kg, single dose, i.p.	↑ Epididymides weight (200 and 500 mg/kg/day) ↑ Sperm count ↓ Abnormal sperm cells (500 mg/kg/day) ↑ SOD activity in plasma (500 mg/kg/day) ↑ CAT and GPx activity in testicular tissues ↓ MDA level in sperm, plasma, and testicular tissues ↓ ROS levels in sperm (500 mg/kg/day) ↓ O2 ⁻ levels in sperm (200 and 500 mg/kg/day) ↓ NO production in sperm ↑ MMP in sperm (200 and 500 mg/kg/day) ↓ Testosterone level in serum	[80]
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(Continues)

TABLE 2 (Continued)

Natural agents	Dose, duration, and route of administration	Animal Species	CP dose, duration, and route of administration	Key findings	Reference
<i>Spirulina platensis</i>	1% powder, for 4 weeks, orally (mixed with diet)	Albino rats	3.3 mg/kg/day, for 3 days, i.p.	↑ Epididymal alpha-glucosidase activities, amelioration of seminiferous tubules morphology ↑ Body and testicular weight ↑ Diameters of seminiferous tubules and the germinal epithelial height ↑ Serum testosterone level ↓ Atrophic and degenerative tubules ↑ Spermatogenic cells ↓ Expression of Bax proapoptotic protein	[81]
Omega3 fatty acids	300 mg/kg/day for 9 weeks, orally (PO)	Wistar rats	1.5 mg/kg on days 1–5 of each week for 9 weeks, i.p.	↑ Body and testicular weight ↑ Sperm motility and concentration ↓ Sperm abnormal morphology ↓ CMA3 positivity and protamine deficiency in sperm ↓ Ubiquitinated sperm	[82]
Fish oil	1 softgel/rat/day, for 14 days, orally (PO)	Sprague Dawley (SD) rats	7 mg/kg, single dose, i.p.	↓ Lipid peroxidation in testicular tissue ↑ CAT, SOD, GPx, and GSH-Px activity in testicular tissues - No effect on testicular and other reproductive organ weights ↑ Sperm count and motility ↓ Sperm abnormality ↓ Histopathological alterations ↑ MSTD and GECT ↓ Atrophic and degenerative tubules	[83]
Honey	5 mL/kg/day, for 5 days, orally (PO)	Swiss albino mice	20, 40, or 100 mg/m ² /day, for 5 days, i.p.	↓ The number of abnormal sperm heads	[84]
Royal Jelly	50 and 100 mg/kg/day, for 10 days, orally (PO)	Wistar albino rats	7 mg/kg, single dose, i.p.	↑ Testicular weights and dimensions and other reproductive organ weights ↑ Sperm concentration and sperm motility	[85]

TABLE 2 (Continued)

Natural agents	Dose, duration, and route of administration	Animal Species	CP dose, duration, and route of administration	Key findings	Reference
				↓ Testes degeneration, necrosis, and interstitial edema ↓ MDA concentration in testicular tissues ↑ CAT, SOD, and GSH-px level in testicular tissues	

Abbreviations: 8-OHdG, 8-hydroxy-2'-deoxyguanosine; cDDP, cisplatin; AHE, acacia hydaspica ethyl acetate; AsA, ascorbic acid; AST, aspartate amino transferase; ATM, ataxia telangiectasia mutated; BAX, BCL2-associated X protein; Bcl-2, B-cell lymphoma 2; BUN, blood urea nitrogen; CAT, catalase; CMA3, chromomycin A3 staining; COX-2, cyclooxygenase-2; CYP11A1, P450 side chain cleavage enzyme; CYP17A1, 17 α -hydroxylase; FSH, follicle stimulating hormone; GECT, germinalepithelialcellthickness; GR, glutathione reductase; GRP-78, glucose-regulated protein-78; GSH, glutathione; GSH-Px, glutathione peroxidase; GSSG, oxidized glutathione; GST, glutathione S-transferase; HSD, hydroxysteroid dehydrogenase; HSD17B, 17 β -hydroxysteroid dehydrogenase; HSD3B1, 3 β -hydroxysteroid dehydrogenase; IL-1 β , interleukin-1 β ; IL-6, interleukin-6; iNOS, inducible nitric oxide synthase; LH, luteinizing hormone; Lhcgr, luteinizing hormone receptor; LPO, lipid peroxides; MAPK, mitogen-activated protein kinase; MDA, malondialdehyde; MMP, mitochondrial membrane potential; MPO, myeloperoxidase; MSTD, mean seminiferous tubule diameter; NF- κ B, nuclear factor-kappa B; NO, nitric oxide; OSI, oxidative stress index; PCNA, proliferating cell nuclear antigen; p-IRE1, phosphorylated inositol-requiring transmembrane kinase/endonuclease 1; p-JNK, phosphorylated c-jun-N-terminal kinase; POD, peroxidase; QR, quinone reductase; ROS, reactive oxygen species; Scarb1, lipoprotein-receptor; SOD, superoxide dismutase; STAR, steroidogenic acute regulatory protein; TAC, total antioxidant capacity; T-AOC, total antioxidant capacity; TAS, total antioxidant status; TBARS, thiobarbituric acid reactive substances; TNOS, total nitric oxide synthases; TOS, total oxidant status; TUNEL, transferase dUTP nickend labeling; XO, xanthine oxidase; γ -GT, gamma glutamyl transpeptidase; γ -H2AX, H2A histone family member X.

CP three times a week for 35 days at the dose of 3 mg/kg while DPP and *P. ginseng* extracts were administered with two different daily doses of 200 and 400 mg/kg for 8 weeks by oral gavage. These herbal extracts positively affected fertility and testicular histopathological changes caused by CP. Sperm parameters were assessed by using DPP and *P. ginseng* extracts in CP-treated rats and resulted in an increase in sperm count, motility, and viability and also a decrease in sperm abnormalities only by using 400 mg/kg extracts. Moreover, serum levels of testosterone and testicular GSH activity were elevated, and in contrast testicular levels of MDA decreased.^[29]

3.10 | *Diospyros lotus* L.

Different parts of the *D. lotus* tree have traditionally been known for various medicinal uses around the world. Its fruits are carminative, astringent and the seeds are sedative, astringent, and febrifuge.^[96] The fruits extract has been demonstrated with beneficial effects on testicular toxicity caused by CP. It was administered orally at the dose of 1000 mg/kg/day for 10 days in combination with an intraperitoneal single dose of 7 mg/kg CP. The results show that testicular toxicity is significantly reversed in terms of increase in sperm concentration and motility and decrease in sperm abnormalities. *Diospyros lotus* elevated testosterone concentration in plasma and enhanced antioxidant capacity by increasing in SOD and GSH amounts in testicular tissue. As an apoptotic marker, the density of caspase-3 was significantly inhibited by the administration of *D. lotus*.^[30]

3.11 | Ellagic acid

Ellagic acid is a natural phenolic lactone constituent extracted from numerous fruits, seeds, and vegetables with the highest concentration found in the fruit of *Rubus idaeus* (raspberries).^[97] Ellagic acid has shown a wide diversity of biological activities including anti-inflammatory, antioxidant, anti-mutagen, estrogenic, and/or anti-estrogenic (estrogen receptor modulator) properties.^[98] Oral administration of ellagic acid at the dose of 10 mg/kg/day for 10 days significantly improved the antioxidant status, histopathological changes, and sperm parameters by reducing oxidative stress in CP-treated rats received an intraperitoneal single dose of 7 mg/kg.^[10] However, the possible molecular mechanism of the protective effect of ellagic acid against CP-induced testicular toxicity requires further investigation to be fully understood. The recent study of Goyal et al. was performed to evaluate the protective effect of ellagic acid against gonadotoxicity caused by cisplatin in mice. Oral administration of ellagic acid (50 mg/kg/day for 6 weeks) markedly ameliorated reproductive functions and decreased cisplatin-induced toxicity (5 mg/kg). Ellagic acid led to positive histopathological changes in the testis and a significant reduction in the rate of peroxidative tissue damage.^[31]

3.12 | Epigallocatechin-3-gallate

Epigallocatechin-3-gallate is a polyphenolic flavonoid and is one of the essential constituents found in green tea. Epigallocatechin-3-gallate is a known free radical scavenger, anti-inflammatory, and anticancer substance. Intraperitoneal epigallocatechin-3-gallate at doses of 40 and 80 mg/kg/day, for 5 days provided remarkable prevention against testicular damage caused by intraperitoneal 10 mg/kg single dose of CP in Sprague Dawley rats. Biochemical measures indicated that epigallocatechin-3-gallate increased serum testosterone level and testicular total antioxidant status and decreased oxidative stress and inflammatory mediators including MDA, NO, interleukin 6 (IL-6), and IL-1 β in testicular tissue. Cytochrome C, Bax/Bcl-2 ratio, and caspase-3, the apoptotic biomarkers were also significantly attenuated in epigallocatechin-3-gallate treated rats compared to CP-administered rats. Moreover, histopathological findings of rat testes showed that testicular injury scores subsided. Epigallocatechin-3-gallate also enhanced spermatogenesis; however, this effect was only observed with the dose of 80 mg/kg epigallocatechin-3-gallate. Finally, the expression of TNF- α as an inflammatory factor in testes was measured by immunohistochemistry method, which showed that epigallocatechin-3-gallate reversed elevated level of TNF- α induced by CP.^[32]

3.13 | Eugenol

4-Allyl-1-hydroxy-2-methoxybenzene (eugenol) is the main component of the oil of clove plant (*Eugenia caryophyllata*), which has effects of antioxidant, antimicrobial, antiviral, pain killer, anti-inflammatory, anticarcinogenic, and apoptosis-promoting. The eugenol's antioxidant effect is due to its methoxy-phenolic structure. Several animal studies have shown the antioxidant activity of eugenol that remarkably carried out protection against oxidative tissue damage. Eugenol is nontoxic, noncarcinogenic, and nonmutagenic and according to the United States, Food and Drug Administration (FDA) is considered safe.^[99,100] An intraperitoneal single dose of cisplatin (15 mg/kg) was given to rats. Eugenol treatment (100 mg/kg/day for 5 days, i.p.), decreased the severity of cisplatin-induced histopathological damage of testes, and improved sperm motility. Also, the GSH, GPx, SOD, and CAT activities increased and lipid peroxidation activities reduced.^[33]

3.14 | *Eugenia jambolana* Lam.

Eugenia jambolana is a tropical tree belonging to the family Myrtaceae. The fruit of the plant, commonly known as Jamun, has been demonstrated to possess antioxidant, anti-inflammatory, antidiabetic, hepatoprotective, antibacterial, and anticancer activities.^[101] Anand et al. showed that treatment with *E. jambolana* extract at the dose of 25 mg/kg provided significant rescue of testicular damage caused by CP in rats.^[34] By virtue of its strong antioxidant and antiapoptotic activity, *E. jambolana* could be considered as a compound with a potential protective effect against CP-induced testicular toxicity for further studies.

3.15 | *Ficus carica* L.

F. carica Linn (Moraceae) is an important plant in the genus *Ficus* and is commonly referred to as "fig." It is one of the first plants that were cultivated in the world. Phytochemical studies on fruits and leaves of fig have demonstrated that they contain large amounts of volatile compounds, phenolics, and organic acids. Extracts from different parts were shown to possess a wide range of therapeutic effects including immunosuppressive, anti-inflammatory, anticancer, hypoglycemic, and hepatoprotective effects.^[35,102] A study in 2020 proved the protective potential of methanol extract of *F. carica* leaves on CP-induced genotoxicity and testicular damage (10 and 15 mg/kg). *F. carica*, potentially due to its richness in flavonoids and phenolic compounds, reduced the CP-induced testicular damage. Stimulation of TNF- α gene expression and histopathological alterations in the testis were decreased after the fig extract supplementation. Also, *F. carica* at different doses (cotreatment with fig and cisplatin) alleviated the overexpression of the iNOS gene compared to cisplatin alone.^[35]

3.16 | *Fraxinus xanthoxyloides* (G. Don) wall. ex A. DC

Fraxinus xanthoxyloides commonly known as Afghan ash is a medicinal plant of family Oleaceae. It is a potential source for steroids, flavonoids, phenylethanoids, lignins, coumarins, and terpenoids. Administration of *F. xanthoxyloides* (200 and 400 mg/kg/day) for 45 days, after a single dose of CP (10 mg/kg) proved the potential efficacy of the leave extract against CP-induced reproductive toxicity in rats. Ijaz et al. reported that the *F. xanthoxyloides* treatment elevated hormone concentrations (FSH, LH, and testosterone), germ cell numbers, and testicular antioxidant capacity.^[36]

3.17 | *Ginkgo biloba* L.

G. biloba is a well-known medicinal plant used in traditional Chinese medicine and has a long history of therapeutic uses. Its leaves have been widely shown to have various pharmacological properties such as antioxidant, anti-inflammatory, neuroprotective, nootropic, and anticancer effects.^[103] Its main bioactive constituents are terpenoids, flavonol glycosides, and proanthocyanidins. Flavonol glycosides (quercetin, catechin) are the most abundant among these three groups. The terpenoids (ginkgolides and bilobalides), are major components of *G. biloba*.^[104] Oral administration of *G. biloba* leaf extract has been shown to attenuate the CP-induced toxicity in the reproductive system of rats. It has been shown that it restores the weights of testes and epididymis, normalizes sperm count and motility, and alleviates the increased germ-cell degeneration in seminiferous tubules and germ-cell apoptosis. It also decreased the testicular MDA levels and MPO activity, increased the SOD and CAT activities in testes, and downregulated the COX-2, iNOS, and nuclear factor-kappa B (NF- κ B) p65 in testicular tissues.^[37]

3.18 | Grape seed

Studies have shown that grape seed procyanidin extract (GSPE) can ameliorate testicular damage of cisplatin by inhibiting free radicals in animals. In 2018, Tian et al. explored the antagonistic mechanisms of GSPE against testicular damage caused by cisplatin administration. A total of 100, 200, and 400 mg/kg/day of GSPE were given through gavage for 15 days consecutively to rats that were received 10 mg/kg single dose of CP. The results demonstrated that GSPE ameliorated the testicular pathological alterations induced by CP and increased body and testes weights and serum level of testosterone. GSPE raised the mRNA expression and enzymes of testosterone synthesis levels and also diminished CP-induced oxidative/nitrative stress in testicles of rat.^[39]

3.19 | *Ichnocarpus frutescens* (L.) W.T. Aiton

In a model of CP-induced testicular toxicity (10 mg/kg, single dose, p.o.), *Ichnocarpus frutescens* (aqueous root extract) and *Cyperus rotundus* (aqueous rhizome extract) were administered in three different doses (100, 200, and 400 mg/kg, single dose, p.o.). It has been shown that all doses of both extracts improved gonadotoxicity measures. Oral administration of mentioned extracts increased the number of spermatids and spermatozoa, sperm density and motility, and testosterone production. It also altered the CP-induced changes in MDA, CAT, and GSH levels in testicular tissues. *I. frutescens* and *C. rotundus* extracts elevated testosterone, FSH, and LH blood levels compared with CP-treated rats. The protective effects of these extracts were dose-dependent. It was shown that the most effective dose was 400 mg/kg. The mechanism of action of these extracts is probably prevention of free radical production and promotion of antioxidant defense in testicular tissues which eventually improves testes functionality.^[42]

3.20 | *Launaea taraxacifolia* (Willd.) amin ex C. Jeffrey

The leaves of *L. taraxacifolia* (Asteraceae) have been known for various medicinal uses around the world. The extract has been demonstrated to have neuroprotective and hepatoprotective effects in the cisplatin toxicity models.^[105] Furthermore, oral administration of *L. taraxacifolia* at the dose of 100 and 400 mg/kg/day for 21 days significantly reversed CP-induced testicular toxicity in terms of sperm count and motility improvement and decrease in sperm abnormalities. *L. taraxacifolia* augmented oxidant/antioxidant status by increasing CAT and SOD activity and glutathione levels in testicular tissue.^[45]

3.21 | Lycopene

Lycopene is a kind of carotenoid with significant antioxidant activities. The effects of lycopene were assessed on CP-induced sperm toxicity, which has been reported to be effective. CP-injected rats

(7 mg/kg; i.p.) were treated by oral administration of lycopene in two different ways and received lycopene for 10 days before or for 5 days after CP injection. Decrease in sperm count and motility, as well as sperm abnormalities induced by CP, were resolved with posttreatment of lycopene. In addition, testicular lipid peroxidation parameters such as MDA and GSH, restored by lycopene administration. In general, administration of lycopene after CP injection resulted in a higher level of protection compared with lycopene pretreatment.^[46]

3.22 | *Maclura pomifera* (Raf.) C.K. Schneid

Maclura pomifera (Raf.) Schneid. commonly known as the Osage orange, is a tree of the mulberry family (Moraceae). Several studies showed that *M. pomifera* (MP) and its bioactive components have pharmacological properties as an anticancer, anti-inflammatory, analgesic, antibacterial, and antidiabetic. The effect of MP extract on attenuating the reproductive toxicity of cisplatin has been reported by Aksu et al.^[48]

3.23 | *Melissa officinalis* L.

Administration of *M. officinalis* (1000 mg/kg/day) and vitamin E (100 mg/kg/day) for 7 days, after a single dose of CP (10 mg/kg) reversed spermiotoxicity. In fact, it was reported that the treatment improved testes' weight, sperm count, motility and viability, plasma levels of testosterone, FSH, and LH.^[106]

3.24 | Motiliperm (*Morinda officinalis* F.C. How., *Cuscuta chinensis* Lam. and *Allium cepa* L.)

Motiliperm is a herbal compound that is still under investigation by Dong-A pharmaceutical company and consists of *M. officinalis*, *C. chinensis*, and *A. cepa* with beneficial effects on male fertility. The effects of orally administered 100 and 200 mg/kg/day Motiliperm for 28 days were evaluated on testicular toxicity by a single dose of CP in rats. Administration of Motiliperm decreased proteins involved in apoptotic endoplasmic reticulum stress signaling pathway in testicular tissues including glucose-regulated protein-78 (GRP-78), phosphorylated inositol-requiring transmembrane kinase/endoribonuclease 1 (p-IRE1), and phosphorylated c-jun-N-terminal kinase (p-JNK). Apoptotic sperm cells—measured by using seminal fluid flow cytometry—remarkably decreased with Motiliperm. Johnsen's score—which is determined based on the amount of spermatogenic cells in seminiferous tubules—is increased by administration of 200 mg/kg Motiliperm in CP-treated rats. Decreased epididymis and prostate weight caused by CP improved, respectively, by 100 and 200 mg/kg Motiliperm administration compared to CP, while body weight was increased by using both doses of 100, 200 mg/kg Motiliperm compared with CP-treated rats. Motiliperm and CP cotreatment of rats decreased testicular level of MDA with both doses of Motiliperm and increased blood concentration of testosterone

only with Motiliperm 100 mg/kg compared to CP. However, any significant effect was not observed on testicular weight, number and motility of sperm, spermatogenic cell density, and content of steroidogenic acute regulatory (StAR) protein compared to a single administration of CP.^[50]

3.25 | Naringenin (NG)

NG is proven to have anti-inflammatory, antioxidant, anti-nitrative, and antiapoptotic activities. NG prevented CP-induced gonadotoxicity in Sprague Dawley rats. The rats received one injection of CP (10 mg/kg) and 10 days orally NG (50 mg/kg/day). NG remarkably raised testosterone of serum, increased total antioxidant status, and the Bcl-2/Bax ratio in testes. In addition, reduced caspase-3, TNF- α /IL-10 ratio and the elevated MDA level in testicular tissue in CP-challenged rats. NG also meaningfully increased P-glycoprotein level and reduced histopathological alterations and testicular injury score and immunohistochemical activity of iNOS induced by cisplatin in testes of rats.^[51]

3.26 | *Nigella sativa* L.

Various biological activities of *N. sativa* have been widely studied. Thymoquinone—a chemical substance in *N. sativa* oil—is responsible for most pharmacological properties of this herb, for instance, reduction in inflammatory cytokines, oxidative damages, lipid peroxidation, and carcinogenesis.^[107] In a study, Wistar albino rats were treated with *N. sativa* oil and CP simultaneously, with the daily dose of 0.5 ml/kg and 1 mg/kg, respectively. Both substances were administered (i.p.) for 21 days. Results showed that *N. sativa* oil has protective effects by normalizing MDA concentration. It also reversed seminiferous tubules atrophy, testicular degeneration, and necrosis.^[52]

3.27 | Nucleosides-rich extract from *Cordyceps cicadae*

Prophylactic oral administration (50, 100 and 400 mg/kg/day) of nucleosides-rich extract from *Cordyceps cicadae* (NRCE) for 7 days before cisplatin-induced testicular damage (10 mg/kg, i.p.) suppressed pathological changes caused by cisplatin administration and raised the antioxidant and anti-inflammatory activities in testes.^[24]

3.28 | Olive

The leaves of *Olea europaea* (Olive) are rich in polyphenolic and flavonoid compounds that were demonstrated to possess anti-inflammatory, antioxidant, and anticancer effects.^[108,109] In a study, the effect of oral administration of olive leaf extract (OLE)

(300 mg/kg/day) for 5 consecutive days against testicular damage caused by an intraperitoneal injection of CP at a dose of 5 mg/kg in Wistar rats, was tested. OLE elevated body and testicular weight, serum testosterone, LH, and FSH concentration as well as antioxidant enzymes (CAT, SOD, and GSH enzyme) in testicular tissues. Additionally, OLE significantly decreased MDA, nitrite/nitrate, IL-1 β , and TNF- α levels in testicular tissue. Besides this, it diminished the histopathological changes and raised mRNA expression levels of Nrf2 and HO-1 pathway.^[53]

3.29 | Pomegranate

Different parts of pomegranate fruit have strong antioxidant activity. Also, its flowers have high antioxidant activity. The antioxidant potential of pomegranate juice is due to its high content of polyphenolic compounds including ellagic acid and ellagitannins.^[110] In an animal study aimed at evaluating the protective effect of pomegranate seed extract on oxidative stress induced by cisplatin in rabbit testis, it was shown that administration of pomegranate seed extract (250 mg/kg; p.o., for 6 days before/after a single dose of 5 mg/kg cisplatin; i.p.) reduced cisplatin-induced elevated levels of malondialdehyde. The extract also increased catalase, glutathione peroxidase, and glutathione levels in rabbit testicular tissue.^[54] This study suggested that pomegranate seed extract supplementation may have a protective role in cisplatin-induced testicular toxicity.

3.30 | *Prosopis cineraria* (L.) Druce

The unripe fruit pod of *P. cineraria* is claimed to have extraordinary effects in the treatment of male infertility and increase the volume and count of sperm. According to the results of a 2020 study, doses of 400 and 800 mg/kg/day of the hydroalcoholic extract of *P. cineraria* orally improve cisplatin-induced testicular toxicity. *P. cineraria* dramatically elevated body and testis weight as well as sperm count, motility, and viability. Moreover, it increased LH, FSH, and testosterone levels in serum and SOD, CAT, LPO, GPx, and GSH in testicular tissues.^[55]

3.31 | Quercetin

Quercetin is a flavonoid that is found in some fruits and vegetables and it has been studied as a preventive agent against oxidative stress reactions. The effects of quercetin on CP-related testicular toxicity were assessed which shows testicular damage measures are partly improved by quercetin; 50 mg/kg/day quercetin was given for 10 days through gavage to the male rats that were additionally received 7 mg/kg i.p. single dose of CP in two different ways, namely quercetin before CP and CP before quercetin. Administration of quercetin before CP reduced histopathological changes in testicular tissue while quercetin had no significant effect on testicular total

antioxidant capacity and total oxidant status. In fact, quercetin did not affect testicular levels of MDA, catalase, SOD, and xanthine oxidase.^[56]

3.32 | Resveratrol

Resveratrol (3,5,4'-trihydroxy-trans-stilbene) is a natural phytoalexin and a polyphenolic compound found in many plants including grapes (*Vitis vinifera*), peanuts, berries, and roots of polygonum cuspidatum (*Japanese knotweed*).^[111] The biomedical and pharmacological potential of resveratrol has been widely reported and its protective effects against CP-induced toxicity in various organs such as the heart, kidney, brain, and ear have been studied.^[58] It has been demonstrated that resveratrol attenuated CP-induced apoptosis and genotoxicity in somatic and germinal cells by augmenting the antioxidant defense system. Pretreatment with oral resveratrol (50, 100 mg/kg/day) for 7–14 days before single dose of intraperitoneal CP resulted in a significant, dose-dependent decrease in genotoxic changes such as DNA strand-breaks, spermatogonial chromosomal aberrations, and bone-marrow suppression.^[57] By virtue of its steroidogenic and antioxidant activity, resveratrol improved testicular and epididymal toxicity induced by CP in rats. Intraperitoneal administration of resveratrol (20 mg/kg/day) for 45 days could elevate sperm parameters and antioxidant enzyme activities in the testicular and epididymal tissues. It also increased testicular steroidogenic enzymes activities in testes and testosterone level in circulation.^[12] Similarly, another study reported that oral administration of resveratrol (1 mg/kg) for 28 days, elicited protective action by mitigating testicular oxidative stress.^[58] Also, the results of a study in 2018 showed that resveratrol protected testicular damage and genotoxicity caused by cisplatin. It increased antioxidant enzyme activities, testosterone, FSH, and LH levels in serum, increased body and testis weights as well as sperm count and motility. In addition, it decreased testicular apoptosis and inhibited the expression of cleaved caspase-3, P53, JNK, ERK1/2.^[59] The effects of resveratrol on CP-induced reproductive toxicity have been shown in more detail in Table 1.

3.33 | *Rhodiola rosea* L.

Rhodiola is a plant genus with a wide variety of therapeutic effects. *Rhodiola rosea* L. extract contains two main components including salidroside and tyrosol which contribute to the majority of its effects. Studies demonstrated *R. rosea* extract exhibited antioxidant, anti-inflammatory, anticancer, anti-nociception, and immune system modification activity. Moreover, *R. rosea* extract showed preventive effects against oxidative stress induced by cardiovascular, diabetes mellitus, liver, skin, and neurodegenerative diseases.^[112] The effect of *R. Rosea* extract (0.0125–2.5 mg/L) was evaluated in CP-treated (0.0147 g/L) mouse TM4 Sertoli cell line by using MTT assay. *R. rosea* extracts provided a protective effect against CP toxicity in the Sertoli

cell line through an increase in TM4 cell line proliferation and amelioration of morphological deformation. In addition, *R. rosea* extracts at the concentration of 0.1 mg/L affected antioxidant markers, that is, it decreased the amount of MDA and increased the activity of SOD and GSH compared to CP-treated TM4 Sertoli cell line.^[60]

3.34 | Rutin

Rutin (3,3,4',5,7,'-pentahydroxyflavone-3-rhamnoglucoside) is a well-known flavonoid glycoside naturally found in many plants such as buckwheat, passionflower, spinach, onions, apples, and tea.^[61] Various pharmacological activities have been shown by this flavonoid including antioxidant, anti-inflammatory, antiangiogenic, and antiviral properties as well as immunomodulator, antibacterial, antidiarrhoeal, antiulcer, antimutagenic, hepatoprotective, myocardial protecting, and antitumor activities.^[113] Rutin has been indicated to mitigate the CP-induced reproductive toxicity in rats. Rutin treatment with oral administration of 150 mg/kg for 14 days ameliorated the spermatological, biochemical, and histological parameters in Wistar albino rats who received a single dose of 10 mg/kg i.p. CP. Sperm motility, cauda epididymal sperm density, dead sperm percentage, and morphological sperm abnormalities, MDA, GSH, GSH-Px, SOD, and CAT were recovered by rutin. It mitigated the CP-damaged testicles, oxidative stress, and testicular degeneration, and apoptosis.^[61] The results of another study showed the protective effects of rutin against testicular damage of cisplatin in Sprague Dawley rats. Rutin (75 mg/kg) was administrated orally for the next 13 days and cisplatin was given 7 mg/kg (i.p.) on day first and then reproductive organs were analyzed for various parameters. Rutin caused the remarkable increase of intratesticular testosterone concentration, testicular daily sperm production, % DNA in head and head length, germ cell number, epithelial cell height, and CAT, SOD, POD, and glutathione level in testicular tissues, besides, it decreased TBARS and oxidative stress in testicular tissues.^[62]

3.35 | *Satureja hortensis* L.

Phytochemical studies showed that *Satureja hortensis* L. (Lamiaceae) is rich in volatile oil, sterols, tannins, and phenolic compounds.^[114] UHPLC/DAD/HESI-MS/MS analysis demonstrated that *S. hortensis* extract contains high amounts of phenolic compounds with rosmarinic acid as the principal compound, caffeic acid, and NG. A study was designed to investigate the protective effect of the methanolic extract of aerial parts of *S. hortensis* on testicular oxidative damage caused by cisplatin administration. Wistar rats received *S. hortensis* orally at doses of 50, 100, and 200 mg/kg for 10 days. Cisplatin was given 7.5 mg/kg intraperitoneally on the fifth day of the experiment. Extract of *S. hortensis* increased levels of serum parameters like testosterone, improved tissue oxidative stress parameters, and tissue morphology, also, elevated Bcl-2/Bax ratio as a sign of apoptosis in rats.^[63]

3.36 | Speman®: A herbal formulation

Speman is a natural product formulation produced by Himalaya Drug Company and its formulation contains herbal active ingredients including *Orchis mascula*, *Lactuca serriola*, *Hygrophila spinosa*, *Mucuna pruriens*, *Parmelia perlata*, *Argyrea speciosa*, *Tribulus terrestris*, *Leptadenia reticulata*, and Suvarnavang. Speman had been shown in previous studies to have beneficial effects on human and animal male reproductive function, spermatogenesis, and testicular sex hormone production.^[115] It was also demonstrated that Speman improves male infertility, oligospermia, premature ejaculation, spermatorrhoea, and prostate hypertrophy.^[116] In a study on male mice, gonadotoxicity was induced by injecting CP at the dose of 3 mg/kg/day, intraperitoneally for 3 alternate days. Also, Speman was orally administered for 35 days at a dose of 80 mg/kg/day. The results showed that Speman improved sperm characteristics including concentration and motility. In addition, the results of the hypo-osmotic swelling test indicated an increase in the number of viable sperms with the competent cell membrane by Speman administration. It also increased testes and epididymis weight compared to CP-injected mice. Moreover, Speman administration in CP-treated mice has remarkably increased testicular steroidogenic enzyme activities including 3 β -hydroxysteroid dehydrogenase and 17 α -hydroxysteroid dehydrogenase as well as serum testosterone levels.^[117]

3.37 | Tribulus terrestris L.

In recent studies, *T. terrestris* showed to have several pharmacologic properties including antioxidant, antitumor, cardioprotective, anti-inflammatory, antidiabetic, and enhancer of male physical and sexual function. Male mice were administered intraperitoneal *T. terrestris* hydroalcoholic extract in three different doses of 100, 300, and 500 mg/kg/day for 4 days, and the effect of the extract against testicular toxicity induced by CP (5.5 mg/kg; single dose; intraperitoneal) was evaluated. The weight of the testes and the diameter of seminiferous tubules significantly increased with the dose of 300 mg/kg/day *T. terrestris* hydroalcoholic extract. Furthermore, a decrease in apoptotic cells of seminiferous tubules was observed by administering 300 and 500 mg/kg/day *T. terrestris* hydroalcoholic extract which confirms its antioxidant activity.^[65]

3.38 | Trigonella foenum-graecum L.

Fenugreek (*Trigonella foenum-graecum*) seed extract was given orally to the rats at three different doses of 50, 100, and 200 mg/kg/day, for 5 days. The rats were treated with an intraperitoneal single dose of 10 mg/kg CP which caused testicular damage. Fenugreek seed extract attenuated testicular toxicity of CP indicated in the following observations; testes and epididymides weights, sperm count, and motility increased in dose-dependent manner. Also, histopathological and apoptotic changes in testicular tissue decreased, the activities of

antioxidant enzymes such as SOD and catalase were restored, and myeloperoxidase (MPO) activity and MDA content of testes were reduced. Moreover, fenugreek seed extract decreased testicular expression of inflammatory measures induced by CP, including iNOS and NF- κ B.^[118]

3.39 | Zataria multiflora Boiss.

Zataria multiflora Boiss. (ZM), a plant belonging to the family of Lamiaceae, has numerous pharmacological properties such as antiangiogenesis,^[119] anti-inflammatory,^[120] chemoprotective,^[121] radioprotective,^[122] antibacterial, and anti-antioxidant^[123] activities. The main biologically active components of ZM are polyphenolic compounds, mainly carvacrol and thymol.^[124] Thymol has anti-inflammatory and antioxidant effects suppressing lipid peroxidation and protecting normal organs against oxidative stress and chemotherapy-induced toxicity.^[125] It is shown that administration of ZM at the dose of 200 mg/kg for 7 days by gavage decreased the toxic effects of CP (10 mg/kg) in testicular tissue. ZM meaningfully raised spermatogenesis, serum testosterone level, and sperm parameters (sperm motility, count, and viability rate) also, reduced caspase-3 immunoreactivity, histological changes, and oxidative stress (decreased MDA and PC levels, and increased GSH level). Moreover, ZM treatment caused a significant increase in epithelial thickness, seminiferous tubular diameter, and Johnsen's Score in testicular tissue.^[67]

3.40 | Zingiber officinale Roscoe.

Ginger (*Z. officinale*) rhizome has various properties like antioxidant, anti-inflammatory, antiplatelet, antiemetic, antiulcer, anticancer, anti-hyperlipidemic, and hypoglycemic activities.^[126] Ginger contains polyphenols, flavonoids, and volatile oil and its pharmacological activities are due to these compounds.^[127] Various studies examined the protective effects of ginger against toxicities caused by toxicants.^[128] It was also found that ginger oil protects sex hormones and testicular from dysfunction caused by antiepileptic drugs via antioxidant action in mice.^[129] Fresh ginger juice (5 ml/kg/day) was given orally for 5 days and cisplatin (10 mg/kg, i.p.) was administered a single dose on Day 2. Ginger juice showed anti-inflammatory and antioxidant activities and increased CAT, SOD, GSH, and GPx levels in testicular tissues and levels of serum testosterone, LH, and FSH, while it significantly reduced levels of MDA, IL-1 β , TNF- α , IL-6, NF- κ B, NO, and iNOS.^[69] The effects of *Zingiber officinale* root extract and *Hibiscus sabdariffa* (Roselle) dried flower extract was investigated against CP testicular toxicity. Roselle and Ginger extracts were given orally for 26 days at 1 g/kg/day dose for each extract and a single dose of CP 10 mg/kg injected intraperitoneally to the rats. Findings indicated that *H. sabdariffa* and *Z. officinale* extracts restored testicular histopathological changes and sperm count, motility, and abnormalities. Furthermore, oxidative stress and lipid peroxidation markers in testicular tissue were measured in the presence of both extracts which shows a decrease in MDA levels and an increase in GSH and antioxidant enzymes including catalase and SOD. Both extracts

displayed protective effects against testicular apoptotic cell death induced by CP administration. Moreover, *H. sabdariffa* and *Z. officinale* extract decreased the number of apoptotic cells of seminiferous tubules and the amount of apoptosis in sperms. They also inhibited p53 protein expression as an important protein involved in the apoptotic pathway in seminiferous tubules induced by CP treatment in rats.^[14,71]

4 | SOME NUTRACEUTICALS, NUTRITIONAL, AND NATURAL SUPPLEMENTS

4.1 | All-trans retinoic acid (ATRA)

In 2019, Yucel et al. the effects of ATRA on testicular damage caused by cisplatin were examined. ATRA and CP were given intraperitoneally to rats with the doses of 7.5 mg/kg/day for 10 days and 7 mg/kg single dose respectively and then samples were analyzed. Treatment with ATRA dramatically decreased abnormal sperm count and increased normal sperm count and motility. Also, ATRA remarkably decreased spermatogenesis and alleviated TOS and OSI in testicle tissue, but raised the Johnsen score.^[72]

4.2 | L-Ascorbic acid

L-Ascorbic acid is a scavenger antioxidant that is known to protect spermatogenesis and play an important role in semen integrity and fertility in both men and animals.^[130] It also is known to improve the testis weight in several studies which might be due to its antioxidant activity preventing germ cell death. Administration of L-ascorbic acid (i.p.; 10 mg/kg/day) in adult male BALB/C mice treated with a high dose of intraperitoneal CP (1 and 2.5 mg/kg/day) for two separate rounds (each round was 5 days), recovered the testes lost weight. It also prevented CP-induced cytotoxicity (cell death of primary spermatocytes and round spermatids and decrease in sperm count). Moreover, the CP-induced decrease in sperm motility and increase in sperm abnormalities, as well as histopathological structural changes, were recovered by L-ascorbic acid.^[130] The results of another study showed that once a week intraperitoneal administration of CP and vitamin C for 4 weeks at the doses of 2 and 100 mg/kg, respectively, decreased total peroxides in plasma and increased activity of antioxidant enzymes such as catalase and glutathione S-transferase in testicular tissue.^[75] In addition, oral administration of 100 mg/kg/day vitamin C, 20 min before intraperitoneal CP injection, at the dose of 0.4 mg/kg/day for 60 consecutive days, resulted in improvement in sperm morphological changes caused by CP in Wistar albino rats.^[27]

4.3 | Vitamin E

A study in 2019 was performed to investigate the effects of the vitamin E α -tocopherol isomer on cisplatin-induced testicular

dysfunction in Sprague Dawley rats. It was administered orally (50 and 200 mg/kg/day) for 7 weeks in combination with 5 mg/kg CP intraperitoneal single dose. The results showed that vitamin E 50 and 200 mg/kg remarkably elevated sperm motility and morphology of spermatozoa ($p < 0.05$).^[76]

4.4 | L-Carnitine

In a study, the protective potential of L-carnitine was evaluated against gonadotoxicity in prepubertal and adult sperms caused by cisplatin. L-Carnitine dramatically decreased testes lesion and TUNEL-positive cells count, while PCNA-positive cells count elevated. During the adult period, the number of epididymal sperm and viability was increased in rats that received L-carnitine before prepubertal. L-Carnitine decreased cisplatin-induced spermatogenic and testicular damage in prepubertal rats by inducing germ cell proliferation and preventing apoptosis.^[78]

4.5 | L-Cysteine

Cysteine is an amino acid with a functional group of thiol which is responsible for its antioxidant activity. L-Cysteine and CP were given intraperitoneally to Albino rats with doses of 100 mg/kg and 2 mg/kg once a week for 4 weeks, respectively. L-Cysteine in combination with CP decreased oxidative stress biomarkers such as lipid peroxides, total peroxides, and superoxide anions in testes and plasma. L-Cysteine also increased activities of antioxidant enzymatic defense including catalase and SOD in plasma and testicular tissue of CP-treated rats. Decreased levels of GSH and vitamin E in testicular tissue and plasma caused by CP were elevated by L-cysteine administration. Furthermore, L-cysteine reduced testicular histopathological changes and cell apoptosis caused by CP.^[75]

4.6 | Honey

Oral treatment with honey reduced sperm abnormalities induced by CP in mice. Honey was given orally at the dose of 5 ml/kg/day concomitant with 20, 40, or 100 mg/m²/day intraperitoneal CP for 5 days, which resulted in a decrease in the number of sperm head abnormalities. Honey positively affected sperm head abnormality induced by CP specifically in the spermatozoa, spermatid, and secondary spermatocyte stages of spermatogenesis.^[84]

4.7 | Omega 3 fatty acids

Polyunsaturated fatty acids, such as EPA and DHA, have anti-inflammatory, antioxidant, antiapoptotic effects.^[131] Also, omega 3 acids can reduce tumorigenesis and the levels of TNF and PSA. Additionally, omega 3 acids suppress the generation of ROS that

protect against oxidative damage and lipid peroxidation in testes.^[132,133] Administration of omega 3 after CP demonstrated a remarkable decrease in the mean percentage of sperm ubiquitination compared to the CP group ($p < 0.05$), while sperm protamination elevated as compared to the CP group ($p < 0.001$) in rat sperm. Moreover, omega 3 increased body and testicular weight as well as sperm motility and concentration.^[82]

4.8 | Royal jelly

Royal jelly is a bee product that is produced by the hypopharyngeal gland of worker honey bees. It has significant nutritional values and some other bioactive ingredients which mediate its health benefits. Royal jelly is well-known for protecting the reproductive system against toxins. Studies have shown that royal jelly has spermatogenesis stimulating activity and reversed oxidative injuries in mouse testes via reducing pro-inflammatory cytokines. It also enhanced fertility and reduced oxidative biomarkers in male rabbits.^[134] Oral administration of royal jelly (50 and 100 mg/kg/day, for 10 days) with CP (7 mg/kg, single dose, i.p.) in rats, increased testicular weight, sperm count, sperm motility, as well as CAT, SOD, and GSH-Px enzyme in testicular tissues. Moreover, royal jelly decreased testes degeneration, necrosis, and interstitial edema as well as elevated levels of MDA in testicular tissue which was mediated by CP.^[85] The abovementioned positive effects of royal jelly on CP-induced histopathological and biochemical changes in testicular tissue can result from its antioxidant properties.

4.9 | Fucoxanthin-rich extract from brown algae

Fucoxanthin is a xanthophyll carotenoid extracted from macroalgae and microalgae with the highest concentration found in *Sargassum glaucescens* (Sargassaceae). It has shown a wide range of biological activities, such as antioxidant, antiproliferative, anti-inflammatory, and antiangiogenic properties.^[135] Oral administration of a fucoxanthin-rich extract from brown algae at the dose of 100, 200, and 500 mg/kg/day for 5 days significantly relieved the antioxidant status, sperm parameters, and histopathological changes in CP-treated Syrian hamsters received single dose of 7 mg/kg intraperitoneally.^[80]

4.10 | *Spirulina platensis*

Spirulina platensis is a blue-green alga that has been described as a nutritious food. In one study, the protective effects of *S. platensis* on CP-induced testicular toxicity were investigated. *Spirulina platensis* administration (1% powder, for 4 weeks, mixed with diet) caused a significant increase in CP-induced weight loss in the body and testes as well as a decrease in serum testosterone levels in the studied rats. In a histological investigation, testes of animals in the CP group

showed vacuolation, incision and reduction of spermatogenic cells, interstitial edema, capillary congestion, and Leydig cell degeneration, and *S. platensis* caused decrease in the CP-induced histopathologic alterations. Immunohistochemical examination also showed that the testes of the CP-treated group had increased expression of Bax proapoptotic protein. But the *S. platensis*-treated rats showed a significant decrease in Bax overexpression. Hence, the results showed that *S. platensis* could play a protective role against testicular damage caused by CP.^[81]

5 | DISCUSSION

CP is an anticancer drug that is used for the treatment of a variety of cancers and adversely affects the male reproductive system. Several studies have shown various herbal compounds and natural substances significantly reverse the testicular toxicity of CP. In this review, the protective effects of natural compounds, medicinal herbs, and phytochemicals were assessed against CP reproductive toxicity. Among plant extracts and phytochemicals, ginger and curcumin had the highest number of studies individually or in combination, respectively. They were followed by the largest number of articles on resveratrol and grape seed, as well as *G. biloba*. Of course, in general, there are many studies on the mentioned plants and compounds in different fields.

In most of the obtained studies, testicular and epididymal weights decreased in response to CP administration. Due to CP-induced testicular damage, the functionality of testes including spermatogenesis and testosterone production was disrupted.

Several mechanisms are involved in CP-induced cell death. Most of the studies reviewed only reported the production of ROS by CP which is mainly due to damage to differentiating germ cells in short term. But the long-term effect of CP which results in loss of spermatogonial stem cells and/or their ability to regenerate spermatogenesis should be more studied.

Proliferating spermatogonial cells can be killed at low doses of CP, likely by the cross-links and forming DNA adducts. In many of the studies reviewed, high doses of CP (like 5–10 mg/kg) were used in which oxidative stress appears to be involved and consequently damages the differentiating germ cells and somatic cells. Accordingly, many natural products with antioxidant activity are able to protect against this form of damage.

CP increases the amounts of ROS and subsequently stimulates oxidative stress. This causes DNA damage and activation of the mitochondrial apoptotic pathway through caspase-3 which eventually triggers cell death.

Moreover, CP increases lipid peroxidation in testicular tissue which is shown through elevated levels of testicular MDA. The levels of NO increase in testes of CP-treated animals as a result of oxidative stress and overexpression of iNOS in testicular tissue. On the contrary, GSH which is considered as an intracellular defense against oxidative stress-related damage is reduced in the testes following CP administration. Antioxidant components of the cell are utilized for counteracting the

generated ROS which eventually results in diminishing the total antioxidant capacity of the cell. Inflammatory mediators such as TNF- α , IL-6, and IL-1 β are also increased by CP administration.

In addition to herbal compounds with protective effects, the role of some natural supplements and nutraceuticals in CP reproductive toxicity have also been indicated. Specifically, as reviewed in this study, the beneficial effects of some vitamins and minerals such as α -tocopherol, ascorbic acid, zinc, and selenium, as well as omega-3 fatty acids, and algae extracts have been investigated. These agents seem to restore CP-induced testicular toxicity via their well-known antioxidant potential. However, a key point in much of the reviewed studies is that long-term follow-up data have not been considered.

6 | CONCLUSION

It can be concluded from our review that considering the antioxidant properties of the variety of herbs, phytochemicals, and natural substances, they are effective for the management of CP reproductive toxicity, in vivo. However, more preclinical and clinical studies should be performed in the future to determine the long-term effect and bioavailability of these protective agents against CP testicular toxicity in cancer patients before and after treatment with CP. Also, these studies will determine whether the aforementioned herbs and natural substances can prevent reproductive toxicity induced by CP without disturbing its anticancer activity.

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CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

REFERENCES

- [1] J. D. Kort, M. L. Eisenberg, L. S. Millheiser, L. M. Westphal, *CA Cancer J. Clin.* **2014**, 64(2), 118.
- [2] S. Dasari, P. B. Tchounwou, *Eur. J. Pharmacol.* **2014**, 740, 364.
- [3] C. Fung, J. r. Dinh P, S. Ardeshtir-Rouhani-Fard, K. Schaffer, S. D. Fossa, L. B. Travis, *Adv. Urol.* **2018**, 2018.
- [4] G. H. Heeba, A. A. Hamza, S. O. Hassanin, *Toxicol. Lett.* **2016**, 264, 38.
- [5] R. J. Aitken, S. D. Roman, *Oxid. Med. Cell. Longevity* **2008**, 1(1), 15.
- [6] A. Brozovic, A. Ambriović-Ristov, M. Osmak, *Crit. Rev. Toxicol.* **2010**, 40(4), 347.
- [7] C. P. Saris, P. M. van de Vaart, R. C. Rietbroek, F. Bloramaert, *Carcinogenesis* **1996**, 17(12), 2763.
- [8] G. R. Dohle, *Int. J. Urol.* **2010**, 17(4), 327.
- [9] M. R. Sadeghi, *J. Reprod. Infertil.* **2012**, 13(3), 123.
- [10] G. Türk, A. Ateşşahin, M. Sönmez, A. O. Çeribaşı, A. Yüce, *Fertil. Steril.* **2008**, 89(5), 1474.
- [11] Y. O. Ilbey, E. Ozbek, M. Cekmen, A. Simsek, A. Otunctemur, A. Somay, *Hum. Reprod.* **2009**, 24(7), 1717.
- [12] K. P. Reddy, P. Madhu, P. S. Reddy, *Food Chem. Toxicol.* **2016**, 91, 65.
- [13] A. Ateşşahin, E. Sahna, G. Türk, A. O. Ceribaşı, S. Yilmaz, A. Yüce, O. Bulmuş, *J. Pineal Res.* **2006**, 41(1), 21.
- [14] A. Amin, A. A. Hamza, *Asian J. Androl.* **2006**, 8(5), 607.
- [15] A. F. Khafaga, A. E. Bayad, *Int. J. Pharmacol.* **2016**, 12(4), 340.
- [16] T. Mercantepe, D. Unal, L. Tümkaya, Z. Yazici, *Exp. Ther. Med.* **2018**, 15(4), 3404.
- [17] T. Afsar, S. Razak, M. R. Khan, A. Almajwal, *BMC Cancer* **2017**, 17(1), 883.
- [18] T. Jurić, J. S. Katanić Stanković, G. Rosić, D. Selaković, J. Joksimović, D. Mišić, V. Stanković, V. Mihailović, S. Afr. J. Bot. **2020**, 128, 141.
- [19] I. O. Sherif, A. Abdel-Aziz, O. M. Sarhan, *J. Biochem. Mol. Toxicol.* **2014**, 28(11), 515.
- [20] K. Kaya, O. Ciftci, M. Aydın, A. Cetin, N. Basak, *Andrologia* **2019**, 51, e13342.
- [21] M. Khazaei, P. D. Bayat, A. Ghanbari, S. Khazaei, M. Feizian, A. Khodaei, H.A.S. Alian, *Indian J. Exp. Biol.* **2012**, 50, 638.
- [22] E. H. Aksu, F. M. Kandemir, S. Altun, S. Küçükler, S. Çomaklı, A. D. Ömür, *J. Biochem. Mol. Toxicol.* **2016**, 30(10), 513.
- [23] S. A. Adelakun, B. K. Akinola, G. T. Akingbade, *J. Family Reprod. Health* **2018**, 12(3), 148.
- [24] L. Wang, Y. He, Y. Li, C. Pei, O. J. Olatunji, J. Tang, A. C. Famurewa, H. Wang, B. Yan, *Chem. Biodiversity* **2020**, 17(11), e2000671.
- [25] E. Tousson, E. Hafez, A. Masoud, A. A. Hassan, *J. Cancer Res. Treat.* **2014**, 2(3), 64.
- [26] Y. O. Ilbey, E. Ozbek, M. Cekmen, A. Simsek, A. Otunctemur, A. Somay, *Hum. Reprod.* **2009**, 24(7), 1717.
- [27] S. E. Elballat, *J. Basic Appl. Zool.* **2016**, 76, 52.
- [28] F. Gevrek, F. Erdemir, *Turk. J. Urol.* **2018**, 44(1), 16.
- [29] G. A. Gabr, G. A. Soliman, M. A. Ganaie, H. M. M. Hassan, *Pharm. Allied Sci.* **2014**, 3(5), 605.
- [30] S. Saral, E. Ozcelik, A. Cetin, O. Saral, N. Basak, M. Aydın, O. Ciftci, *Andrologia* **2016**, 48(3), 308.
- [31] Y. Goyal, A. Koul, P. Ranawat, *Mol. Cell. Biochem.* **2019**, 453(1–2), 205.
- [32] A. A. Fouad, H. O. Qutub, A. E. A. Fouad, A. M. Audeh, W. N. Al-Melhim, *Pharm. Biol.* **2017**, 55(1), 1710.
- [33] F. N. Ekinci Akdemir, S. Yildirim, F. M. Kandemir, E. H. Aksu, M. C. Guler, H. Kiziltunc Ozmen, S. Kucukler, G. Eser, *Andrologia* **2019**, 51, e13353.
- [34] H. Anand, M. M. Misro, S. B. Sharma, S. Prakash, *Andrologia* **2015**, 47(2), 194.
- [35] O. Ahmed, U. A. Fahmy, R. Bakhaidar, M. A. El-Moselhy, S. Z. Okbazghi, A. F. Ahmed, A. Hammad, N. A. Alhakamy, *Pharmacogn. J.* **2020**, 12(3), 645.
- [36] M. U. Ijaz, *Pak. Vet. J.* **2020**, 40(4), 489.
- [37] A. Amin, C. Abraham, A. A. Hamza, Z. A. Abdalla, S. B. Al-Shamsi, S. S. Harethi, S. Daoud, *J. Biomed. Biotechnol.* **2012**, 2012, 362049.
- [38] A. Amin, D. Mahmoud-Ghoneim, M. I. Syam, S. Daoud, *Theor. Biol. Med. Model.* **2012**, 9, 1.
- [39] M. Tian, F. Liu, H. Liu, Q. Zhang, L. Li, X. Hou, J. Zhao, S. Li, X. Chang, Y. Sun, *Syst. Biol. Reprod. Med.* **2018**, 64(4), 246.
- [40] Y.-m Zhao, L. P. Gao, H. L. Zhang, J. X. Guo, P. P. Guo, *Food Funct.* **2014**, 5(3), 605.
- [41] N. C. Yildirim, F. M. Kandemir, F. Benzer, *Dig. J. Nanomat. Biostruct.* **2011**, 6, 155.
- [42] J. Anbu, S. Nithya, R. Kannadhasan, G. Kishore, A. Anjana, S. Suganya, *Int. J. Pharm. Pharm. Sci.* **2012**, 4(1), 437.

- [43] S. Elesha, *J. Med. Plants Res.* **2011**, 5(3), 388.
- [44] O. O. Azu, F. I. O. Duru, A. A. Osinubi, A. A. Oremosu, C. C. Noronha, S. O. Elesha, A. O. Okanlawon, *Middle East Fertil. Soc. J.* **2010**, 15(3), 200.
- [45] S. A. Adejuwon, O. M. Femi-Akinlosotu, J. O. Omirinde, *Andrologia* **2015**, 47(5), 553.
- [46] A. Ateşşahin, I. Karahan, G. Türk, S. Gür, S. Yılmaz, A. O. Ceribaşı, *Reprod. Toxicol.* **2006**, 21(1), 42.
- [47] E. A. Salem, N. A. Salem, A. M. Maarouf, E. C. Serefoglu, W. J. G. Hellstrom, *Urology* **2012**, 79(5), 1184.e1–e6.
- [48] E. H. Aksu, O. Akman, M. Özkaraca, A. D. Ömür, O. Ucar, *Kafkas Univ. Vet. Fak. Derg.* **2015**, 21(3), 397.
- [49] M. Iman, M. Araghi, T. Heidari, V. Mohammadi, *Recent Pat. Anti-Cancer Drug Discovery* **2017**, 12(1), 73.
- [50] K. K. Soni, L. T. Zhang, J. H. You, S. W. Lee, C. Y. Kim, W. S. Cui, H. J. Chae, H. K. Kim, J. K. Park, *Cancer Cell. Int.* **2015**, 15, 1.
- [51] A. A. Fouad, M. M. Refaie, M. I. Abdelghany, *Toxicol. Mech. Methods* **2019**, 29(1), 67.
- [52] E. A. Awadalla, *Nutrition* **2012**, 2(4), 265.
- [53] R. S. Almeer, A. E. Abdel, *Oxid. Med. Cell. Longevity* **2018**, 2018, 8487248.
- [54] F. Benzer, F. M. Kandemir, N. C. Yildirim, S. T. Ozan, *Asian J. Chem.* **2011**, 23(7), 3231.
- [55] C. Vellapandian, V. R. Vetrivelan, E. S. Sukumaran, *J. Ethnopharmacol.* **2020**, 261, 113070.
- [56] M. Aldemir, E. Okulu, K. Kösemehmetoğlu, K. Ener, F. Topal, O. Evirgen, E. Gürleyik, A. Avci, *Andrologia* **2014**, 46(10), 1089.
- [57] S. M. Attia, *Mutat. Res.* **2012**, 741(1–2), 22.
- [58] I. Singh, Y. Goyal, P. Ranawat, *Chem.-Biol. Interact.* **2017**, 273, 200.
- [59] A. A. Shati, *Syst. Biol. Reprod. Med.* **2019**, 65(3), 236.
- [60] Z. Y. Guo, L. P. Gao, W. W. Li, *Zhonghua. Nan Ke Xue* **2013**, 19(11), 1027.
- [61] E. H. Aksu, F. M. Kandemir, M. Özkaraca, A. D. Ömür, S. Küçükler, S. Çomaklı, *Andrologia* **2017**, 49(1).
- [62] S. Jahan, A. Munawar, S. Razak, S. Anam, Q. U. Ain, H. Ullah, T. Afsar, M. Abulmeaty, A. Almajwal, *BMC Urol.* **2018**, 18(1), 107.
- [63] T. Boroja, J. Katanić, G. Rosić, D. Selaković, J. Joksimović, D. Mišić, V. Stanković, N. Jovičić, V. Mihailović, *Food Chem. Toxicol.* **2018**, 118, 252.
- [64] P. Reddy, S. Sainath, K. Reddy, T. Sowbhagyamma, B. Girish, *Int. J. Green Pharm.* **2011**, 5(4), 286.
- [65] Z. Keshtmand, A. Ghanbari, M. Khazaei, A. Rabzia, *Int. J. Insect Morphol.* **2015**, 33(1), 279.
- [66] A. Hamza, H. Elwy, A. Badawi, *Andrologia* **2016**, 48(2), 211.
- [67] S. Karimi, S. J. Hosseinimehr, H. R. Mohammadi, A. R. Khalatbary, F. T. Amiri, *Iran. J. Basic Med. Sci.* **2018**, 21(6), 607.
- [68] A. F. Abdel-Mohsen, M. S. Elgendy, D. M. Elmarakby, D. I. Eldsouky, *Egypt. J. Histol.* **2013**, 36(2), 312.
- [69] A. C. Famurewa, C. A. Ekeleme-Egedigwe, C. S. Onwe, U. O. Egedigwe, C. O. Okoro, U. J. Egedigwe, N. T. Asogwa, *Andrologia* **2020**, 52(10), e13786.
- [70] E. Fekry, A. A. Rahman, M. M. Awany, S. Makary, *Ultrastruct. Pathol.* **2019**, 43(1), 66.
- [71] A. Amin, A. A. Hamza, A. Kambal, S. Daoud, *Asian J. Androl.* **2008**, 10(2), 291.
- [72] C. Yucel, F. D. Arslan, S. Ekmekci, V. Ulker, E. Kisa, E. Erdogan Yucel, M. Ucar, Y. O. Ilbey, O. Celik, B. I. Basok, Z. Kozacioglu, *World J. Men's Health* **2019**, 37(2), 249.
- [73] K. Narayana, M. Al-Bader, A. Mousa, K. M. Khan, *Eur. J. Pharmacol.* **2012**, 674(2–3), 207.
- [74] K. Narayana, S. Verghese, S. S. Jacob, *Exp. Toxicol. Pathol.* **2009**, 61(6), 553.
- [75] E. A. Ahmed, H. M. Omar, S. Elghaffar, S. M. Ragb, A. Y. Nasser, *Food Chem. Toxicol.* **2011**, 49(5), 1115.
- [76] D. V. Aditya, T. Djojodimedjo, D. M. Soebadi, *Indonesian J. Urol.* **2019**, 26(1), 53.
- [77] M. Shakibaie, H. Forootanfar, E. Jafari, A. Salimi, M. Doostmohammadi, H. R. Rahimi, *Toxicol. Environ. Chem.* **2020**, 102(7–8), 386.
- [78] O. Yaman, Y. Topcu-Tarlacalisir, *Biotech. Histochem.* **2018**, 93(3), 157.
- [79] A. El-Shafaei, R. Abdelmaksoud, A. Elshorbagy, N. Zahran, R. Elabd, *Andrologia* **2018**, 50(9), e13077.
- [80] P. T. Wang, S. Sudirman, M. C. Hsieh, J. Y. Hu, Z. L. Kong, *Biomed. Pharmacother.* **2020**, 125, 109992.
- [81] D. A. Ali, W. M. Ibrahim, *Arch. Sci.* **2013**, 66(6), 172.
- [82] F. Hashemi, S. Razavi, F. Khadivi, *Nutr. Cancer* **2018**, 70(8), 1308.
- [83] O. Ciftci, A. Cetin, M. Aydin, K. Kaya, F. Oguz, *Andrologia* **2014**, 46(10), 1161.
- [84] S. Quita, *Alexandria J. Agric. Res.* **2009**, 54(2), 61.
- [85] S. Silici, O. Ekmekcioglu, G. Eraslan, A. Demirtas, *Urology* **2009**, 74(3), 545.
- [86] T. Afsar, S. Razak, A. Almajwal, M. R. Khan, *BMC Complement. Altern. Med.* **2018**, 18(1), 49.
- [87] T. Afsar, S. Razak, D. Aldisi, M. Shabbir, A. Almajwal, A. A. Al Kheraif, M. Arshad, *Sci. Rep.* **2021**, 11(1), 1.
- [88] K. Savikin, G. Zdunić, N. Menković, J. Zivković, N. Cuić, M. Tereščenko, D. Bigović, *J. Ethnopharmacol.* **2013**, 146(3), 803.
- [89] A. Trendafilova, M. Todorova, M. Nikolova, A. Gavrilova, A. Vitkova, *Nat. Prod. Commun.* **2011**, 6(12), 1934578X1100601216.
- [90] K. Kaya, O. Ciftci, A. Cetin, M. Tecellioglu, N. Başak, *Acta Cirurgica Brasileira* **2016**, 31(3), 198.
- [91] A. Dietrich-Muszalska, B. Olas, B. Kontek, J. Rabe-Jabłońska, *Int. J. Biol. Macromol.* **2011**, 49(1), 113.
- [92] D. Durairajanayagam, *Arab J. Urol.* **2018**, 16(1), 10.
- [93] Z. E. Suntres, J. Coccimiglio, M. Alipour, *Crit. Rev. Food Sci. Nutr.* **2015**, 55(3), 304.
- [94] C. B. Hollenbach, R. S. Bing, F. P. S. Mello, T. L. Schuch, M. Rodrigues, F. B. Mello, J. Mello, *Acta Sci. Vet.* **2015**, 43, 1295.
- [95] S. J. Hewlings, D. S. Kalman, *Foods* **2017**, 6(10), 92.
- [96] G. Uddin, A. Rauf, B. N. Siddiqui, S. Q. Shah, *Middle-East J. Sci. Res.* **2011**, 10(1), 78.
- [97] J. L. Rios, R. M. Giner, M. Marín, M. C. Recio, *Planta Med.* **2018**, 84, 1068.
- [98] J. Landete, *Food Res. Int.* **2011**, 44(5), 1150.
- [99] I. Al-Sharif, A. Remmal, A. Aboussekhra, *BMC Cancer* **2013**, 13(1), 600.
- [100] M. S. Sakat, K. Kilic, F. Akdemir, S. Yildirim, G. Eser, A. Kiziltunc, *Braz. J. Otorhinolaryngol.* **2019**, 85(6), 766.
- [101] J. Xu, T. Liu, Y. Li, W. Liu, Z. Ding, H. Ma, N. P. Seeram, Y. Mu, X. Huang, L. Li, *Mol. Nutr. Food Res.* **2019**, 63(9), e1801307.
- [102] S. Mawa, K. Husain, I. Jantan, *Evidence-Based Integr. Med.* **2013**, 2013, ID 974256.
- [103] K. M. Nash, Z. A. Shah, *Integr. Med. Insights* **2015**, 10, 1.
- [104] N. Brondino, A. De Silvestri, S. Re, N. Lanati, P. Thiemann, A. Verna, E. Emanuele, P. Politi, *J. Evidence-Based Complementary Altern. Med.*, **2013**, 2013, 915691.
- [105] A. Adejuwon, *Eur. J. Med. Plants* **2014**, 4, 528.
- [106] M. Iman, M. Araghi, T. Heidari, V. Mohammadi, *Recent Pat. Anti-Cancer Drug Discovery* **2017**, 12(1), 73.
- [107] A. Ahmad, A. Husain, M. Mujeeb, S. A. Khan, A. K. Najmi, N. A. Siddique, Z. A. Damanhouri, F. Anwar, *Asian Pac. J. Trop. Biomed.* **2013**, 3(5), 337.
- [108] M. Sarbishegi, E. A. C. Gorgich, O. Khajavi, *Nephrourol. Mon.* **2017**, 9(3), e47127.
- [109] S. Al-Quraishy, M. S. Othman, M. A. Dkhil, A. E. Abdel Moneim, *Biomed. Pharmacother.* **2017**, 91, 338.

- [110] H. Akhavan, M. Barzegar, H. Weidlich, B. F. Zimmermann, *J. Chem.* **2015**, 2015, ID 907101.
- [111] A. Y. Berman, R. A. Motechin, M. Y. Wiesenfeld, M. K. Holz, *npj Precis. Oncol.* **2017**, 1(1), 35.
- [112] H.-M. Chiang, H. C. Chen, C. S. Wu, P. Y. Wu, K. C. Wen, *J. Food Drug Anal.* **2015**, 23(3), 359.
- [113] A. Ganeshpurkar, A. K. Saluja, *Saudi Pharm. J.* **2017**, 25(2), 149.
- [114] B. Tepe, M. Cilkiz, *Pharm. Biol.* **2016**, 54(3), 375.
- [115] P. G. Jayatilak, D. S. Pardani, B. Dattatreya, A. R. Sheth, *Indian J. Exp. Biol.* **1976**, 14(2), 170.
- [116] K. Mukherjee, A. Tripathi, K. S. Kulkarni, *Indian J. Clin. Pract.* **2003**, 13(11), 29.
- [117] P. Reddy, S. Sainath, K. Reddy, T. Sowbhagyamma, B. Girish, *Int. J. Green Pharm.* **2011**, 5(4), 286.
- [118] A. A. Hamza, H. M. Elwy, A. M. Badawi, *Andrologia* **2016**, 48(2), 211.
- [119] A. H. Norooznezhad, M. Keshavarz, F. Norooznezhad, K. Mansouri, *Int. J. Hematol. Oncol. Stem Cell Res.* **2017**, 11(2), 96.
- [120] L. A. Nakhai, A. Mohammadirad, N. Yasa, B. Minaie, S. Nikfar, G. Ghazanfari, M. J. Zamani, G. R. Dehghan, H. R. Jamshidi, V. Bushehri, R. Khorasani, M. Abdollahi, *J. Evidence-Based Complementary Altern. Med.* **2007**, 4(1), 43.
- [121] S. J. Hosseinimehr, S. Ahmadashrafi, F. Naghshvar, A. Ahmadi, S. Ehasnalavi, M. Tanha, *Integr. Cancer Ther.* **2010**, 9(2), 219.
- [122] S. J. Hosseinimehr, A. Mahmoudzadeh, A. Ahmadi, S. A. Ashrafi, N. Shafaghati, N. Hedayati, *Cancer Biother. Radiopharm.* **2011**, 26(3), 325.
- [123] F. Sharififar, M. H. Moshafi, S. H. Mansouri, M. Khodashenas, M. Khoshnoodi, *Food Control* **2007**, 18(7), 800.
- [124] H. Sajed, A. Sahebkar, M. Iranshahi, *J. Ethnopharmacol.* **2013**, 145(3), 686.
- [125] H.-A. Arab, M. Fathi, E. Mortezaei, S. J. Hosseinimehr, *Pharm. Biomed. Res.* **2015**, 1(1), 26.
- [126] G. Oboh, A. J. Akinyemi, A. O. Ademiluyi, *Exp. Toxicol. Pathol.* **2012**, 64(1–2), 31.
- [127] S. V. Khotimchenko, I. M. Yakovleva, *Phytochemistry* **2005**, 66(1), 73.
- [128] Y. A. Reddy, M. Chalamaiyah, B. Ramesh, G. Balaji, P. Indira, *J. Food Sci. Technol.* **2014**, 51(5), 908.
- [129] A. I. El Makawy, F. M. Ibrahim, D. M. Mabrouk, K. A. Ahmed, M. Fawzy Ramadan, *Biomed. Pharmacother.* **2019**, 110, 409.
- [130] K. Narayana, S. Verghese, S. S. Jacob, *Exp. Toxicol. Pathol.* **2009**, 61(6), 553.
- [131] I. Zararsiz, I. Kus, N. Akpolat, A. Songur, M. Ogeturk, M. Sarsilmaz, *Cell Biochem. Funct.* **2006**, 24(3), 237.
- [132] U. Gogus, C. Smith, *Int. J. Food Sci. Technol.* **2010**, 45(3), 417.
- [133] R. Uygur, C. Aktas, F. Tulubas, E. Uygur, M. Kanter, M. Erboga, V. Caglar, B. Topcu, O. A. Ozen, *Andrologia* **2014**, 46(8), 917.
- [134] A. M. El-Hanoun, A. E. Elkomy, W. A. Fares, E. H. Shahien, *World Rabbit Sci.* **2014**, 22(3), 241.
- [135] M. Bae, M. B. Kim, Y. K. Park, J. Y. Lee, *Biochim. Biophys. Acta. Mol. Cell Biol. Lipids* **2020**, 1865(11), 158618.

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